

The Alaoglu–Erdős Integer-Exponent Conjecture

Machine-Checked Partial Results,
Exact Conditional Structure,
and Research Directions

A combined survey and research report on the corpus in

<https://github.com/VladimirReshetnikov/ProveIt>

Bottom line. No unconditional proof of the two-base conjecture is obtained, and the reduction to the four exponentials conjecture explains why none should be expected from routine combination of the present methods. The combined corpus does establish, in kernel-verified Lean: a zero-free region $2^x < 4096$ (any nonintegral solution has $x \geq 12$); transcendence of every hypothetical nonintegral solution; a canonical primitive odd core; a unique least nonintegral solution β whose output pair is affinely primitive; and the exact conditional rigidity theorem

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\},$$

together with logarithmic counting, a linear dyadic-block bound, sharp arbitrary-order progression obstructions, density zero, explicit Diophantine lower bounds, the 3-smooth classification, the exact classification of positive rational auxiliary bases whose x -th powers are algebraic (equivalently rational), the exact divisible-hull classification for all positive algebraic auxiliary bases, the exclusion of algebraic outputs at three multiplicatively independent positive algebraic bases, a transcendental mixed-output half-plane and rank-one algebraic ratio locus for second iterates, and the equivalence with both fractional-part gaps and a localized-radical exclusion. Exact unit-block counts, the sharp quadratic cumulative asymptotic, and block averages for every finite trigonometric polynomial are kernel-verified. The stronger continuous-test-function and interval-counting forms of blockwise Weyl equidistribution are proved here only at the paper level. An independent interval computation excludes all nonintegral candidates with $2^x < 2^{20}$; that computation is explicitly *not* a Lean kernel proof. The irreducible remainder is the open radical assertion: no odd $u > 1$ has $u^{\log_2 3} \in \mathbb{Z}[1/3]$.

This document merges, updates, and extends two earlier AI-assisted reports (Claude Fable 5, Anthropic; GPT-5.6 Pro, OpenAI), against the repository state of the combined-report base `83ef82ab1` and the kernel-checked feature modules listed in Appendix B. All Lean identifiers cited below are the authoritative statements.

August 20, 2026

Abstract

The Alaoglu–Erdős conjecture asks whether a real x satisfying $2^x, 3^x \in \mathbb{Z}$ must be an integer. It is a particularly concrete unresolved instance of the four exponentials problem: the three-base analogue with $5^x \in \mathbb{Z}$ added is a theorem, formalized in the repository by an interpolation-determinant argument, while the two-base statement sits exactly at the open 2×2 transcendence threshold.

This report is the merged and expanded successor of two documents: a survey of the machine-checked partial results and an independent research report containing an audit and new conditional structure theory. It serves four purposes. First, it catalogues the kernel-verified Lean corpus — zero-free regions through $2^x < 4096$, Gelfond–Schneider transcendence restrictions, multiplicative rank at most two, a canonical primitive odd core, an exact conditional solution monoid with least generator, simultaneous primitive normalizations of its two outputs, exact natural- and rational-third-base algebraic-output classifications, the divisible-hull classification for arbitrary positive algebraic bases, the number-field determinant theorem for algebraic third outputs, sharp arbitrary-order progression obstructions, zero density, shrinking-target Diophantine bounds, the 3-smooth classification of auxiliary bases, the four-exponentials reduction, and unconditional equivalences with both fractional-part gaps and a localized-radical exclusion — with the precise Lean identifiers and their verification status. Second, it records the now kernel-verified conditional structure: if a counterexample exists, a canonical primitive odd core w , rational-power index d , and least nonintegral solution $\beta = a + d \log_2 w$ exist, and the full solution set is the free additive monoid $\mathbb{N}_0 + \mathbb{N}_0 \beta$; exact dyadic-block counts, their sharp quadratic asymptotic, and finite-trigonometric-polynomial block averages are kernel-verified. Full equidistribution for arbitrary continuous tests or interval indicators remains a paper corollary. Third, it records the now kernel-verified general higher-difference obstruction for arbitrary progression length and an independent 40-digit interval verification excluding all nonintegral candidates below 2^{20} under a stated software trust boundary. Fourth, it consolidates the research program: the conjecture is kernel-verified equivalent to the radical statement that $u^{\log_2 3} \notin \mathbb{Z}[1/3]$ for every odd $u > 1$. Primitive-base reduction, exact radical degree, a generic rational six-exponentials theorem, denominator-unrestricted rational-third-output determinants, algebraic-third-output determinants under monomial injectivity, a generic positive-real algebraic six-exponentials theorem, and the uniform finite-difference API now have kernel-verified forms; the radical exclusion itself remains the open boundary.

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1 Executive summary

1.1 What is proved, and at what level of trust

The full conjecture remains open. Several routes were pushed to their exact failure points: the three-base determinant method does not shrink to two bases without proving a case of the four exponentials conjecture; equidistribution of multiples does not defeat exponentially shrinking integrality gaps; local curvature obstructions cannot reach the sparse global structure; and every successful reduction terminates at the same transcendence-theoretic wall, now isolated in Conjecture 6.3.

To keep unlike kinds of evidence separate, every important statement in this report carries one of four labels.

[kernel-verified Lean]

A proof source is present in the repository, lies on the audited `ExponentialIdentities.lean` import surface, and has been accepted by the Lean kernel in direct or full builds of the cited modules (toolchain 4.31/4.32 series, Mathlib pinned by the repository manifest). No `sorry`, no extra axioms, no `native_decide`.

[paper proof in this report]

A conventional mathematical proof is supplied in this document. It is intended to be suitable for formalization, but no claim of kernel checking is made.

[interval computation]

A finite statement was checked by outward-rounded interval arithmetic. The script, parameters, worst margins, and manifest hashes are recorded in Section 8 and Appendix D. The trusted base includes CPython and `mpmath`; this is not interchangeable with a Lean certificate.

[classical theorem]

A classical theorem quoted from the literature.

An earlier audit, performed against commit 423ac24, correctly observed that three modules cited by the survey (`FiniteCheck4096`, `OddCore`, `FractionalPartDensity`) were absent from that commit tree. That discrepancy is resolved: the three modules were subsequently committed in 98646b455 and are kernel-verified; the survey’s temporary pending-verification remark was removed in 716617d72. Statements from these modules are therefore labeled [\[kernel-verified Lean\]](#) below.

The current feature wave adds directly kernel-checked structural modules `PrimitiveOddCore`, `LeastSolution`, `ThreeDenominatorNormalization`, `RationalPowerIndex`, `PrimitiveGenerator`, and `OutputNormalization`. The counting and Fourier modules are `ExactCounting`, `QuadraticAsymptotic`, `BlockWeyl`, and `BlockEquidistribution`. The rational-output modules are `RationalSixExponentials`, `RationalThirdBase`, `RationalBaseThird`, `AlgebraicRationalBase`, `AlgebraicBaseUnitPower`, `AlgebraicRankThreeEquivalence`, `CoreIndependence`, and `CanonicalRadicalThirdOutput`. The determinant layer is `AlgebraicIntegerDeterminant`. The algebraic-output bridges are `AlgebraicOutputBridge` and `FiniteAlgebraicOutputBridge`. The generic algebraic determinant is `AlgebraicSixExponentials`; the specialized theorem and its corollaries are in `AlgebraicThirdBase`, `AlgebraicConsequences`, and

AlgebraicOutputLocus; the canonical radical upgrades are CanonicalRadicalAlgebraicOutput and CanonicalRadicalDivisibleHull; the exact field diagnostics are in CanonicalRadicalFieldNorm. The radical modules are RadicalReformulation, PrimitiveRadical, and RadicalDegree, together with the exact valuation restrictions in CanonicalRadicalValuation and CanonicalRadicalSharpness; the progression modules are SharperProgressions, ThirdDifference, and HigherDifference. Their public headline theorems were audited with `##print axioms`; only the standard Mathlib axioms `propxt`, `Classical.choice`, and `Quot.sound` occur.

1.2 Headline results

1. **Verified zero-free region.** If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 4096$, then $x \in \mathbb{Z}$; equivalently every nonintegral solution has $x \geq 12$. [\[kernel-verified Lean\]](#) An independent interval scan extends the finite barrier to $2^x < 2^{20}$, i.e. $x > 20$. [\[interval computation\]](#)
2. **Transcendence.** Every nonintegral solution is transcendental; for x with $2^x \in \mathbb{Z}$, algebraicity of x is *equivalent* to integrality. $\vartheta = \log_2 3$ is transcendental. [\[kernel-verified Lean\]](#)
3. **Structure.** Conditional on a counterexample there is a unique gcd-normalized odd core $w > 1$, every candidate has unique coordinates $2^i w^j$, and the solution set has the exact free-monoid classification $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0 \beta$ with a unique least nonintegral generator. Candidate counts below N are at most $\log_2 N \cdot \log_3 N$, and each dyadic block $[2^T, 2^{T+1})$ holds at most $\log_3(2^{T+1})$ candidates. The exact block count $\lfloor (N+1)/\beta \rfloor$, the limit $C(T)/T^2 \rightarrow 1/(2\beta)$, and the equivalence between the conjecture and subquadratic growth along 2^T are kernel-verified. Exact block averages converge to the constant coefficient for every finite trigonometric polynomial. [\[kernel-verified Lean\]](#) The extension to all continuous tests and interval indicators remains paper-level. [\[paper proof in this report\]](#)
4. **Unconditional gap equivalences.** The conjecture is equivalent to the existence of any nonempty open subinterval of $(0, 1)$ free of fractional parts of nonintegral solutions, and to fractional parts being bounded away from 0, or from 1. [\[kernel-verified Lean\]](#)
5. **Exact rational and algebraic output classification.** For a hypothetical nonintegral solution x , a positive natural base a has $a^x \in \mathbb{Q}$ exactly when $a = 2^u 3^v$ for some $u, v \in \mathbb{N}_0$, and a positive rational base q has $q^x \in \mathbb{Q}$ exactly when $q = 2^i 3^j / (2^k 3^\ell)$ for some $i, j, k, \ell \in \mathbb{N}_0$. In the simultaneous primitive normalization, the odd core has rational β -th power exactly when it is 3, the 3-free core has rational β -th power exactly when it is 2, and these two exceptional cases cannot occur together. The natural-base statement remains exact with “rational” replaced by “algebraic”; for positive rational bases, algebraic outputs occur exactly at rational 2, 3-units, and for arbitrary positive algebraic bases they occur exactly in the radical hull of those units. Outside these loci the output is transcendental. In particular, at least one of $2^{x^2}, 3^{x^2}$ is transcendental, every $(2^i 3^j)^{x^2}$ with $i, j > 0$ is transcendental, and the algebraic ratio-output locus has lattice rank at most one. [\[kernel-verified Lean\]](#)
6. **Algebraic-output reformulations.** Under the two integrality hypotheses, 5^x is algebraic exactly when x is integral; both first iterated outputs are algebraic exactly when x is integral; and 6^{x^2} is algebraic exactly when x is integral. The positive-algebraic output locus has multiplicative rank three exactly when x is integral, and the rational squared-output ratio locus has rank two exactly then as well. Universalizing any one of these pointwise statements gives a kernel-verified equivalent formulation of the full Alaoglu–Erdős conjecture. [\[kernel-verified Lean\]](#)

7. **Sharper local combinatorics.** For every $r \geq 2$, the sharp criterion $|\vartheta(\vartheta-1) \cdots (\vartheta-r+1)|h^r < n^{r-\vartheta}$ forbids $r+1$ candidates in arithmetic progression. The underlying arbitrary-order forward-difference mean-value identity is formalized as well, and for the three-term case a rational effective criterion $h^{400} \leq n^{83}$ (step exponent $83/400 = 0.2075$) is verified independently. [\[kernel-verified Lean\]](#)
8. **The irreducible remainder.** The conjecture is equivalent to: no odd integer $u > 1$ satisfies $u^\vartheta \in \mathbb{Z}[1/3]$, with a symmetric base-swapped formulation and an equivalent reduction to canonical power-primitive odd bases. The least radical index gives an irreducible binomial and the exact algebraic degree. [\[kernel-verified Lean\]](#) The exclusion itself remains open even for $u = 3$.

2 The problem and its equivalent forms

2.1 Original statement

Conjecture 2.1 (Alaoglu–Erdős, two-base integer form). *For every $x \in \mathbb{R}$,*

$$2^x \in \mathbb{Z} \quad \text{and} \quad 3^x \in \mathbb{Z} \quad \implies \quad x \in \mathbb{Z}.$$

Status: recorded in Lean as the proposition `AlaogluErdosConjecture`, without being asserted.

Since 2^x and 3^x are positive, the hypotheses actually put them in $\mathbb{N}$; and $2^x \in \mathbb{N}$ forces $x \geq 0$, since a negative exponent gives $0 < 2^x < 1$. All solutions therefore live in $[0, \infty)$ (`nonneg_of_two_rpow_integer`).

Questions of this type go back to the 1944 study of colossally abundant numbers by Alaoglu and Erdős [1], where the arithmetic nature of simultaneous rational powers p^x, q^x of distinct primes controls the quotients of consecutive colossally abundant numbers — a theme reaching back to Ramanujan’s superior highly composite numbers [2]. The modern two-base integer form is elementary to state and is a direct special case of the four exponentials conjecture (Schneider’s first problem [7]; see Waldschmidt [8, 11]). As of the dated literature check for this report, the four exponentials conjecture remains open.

2.2 Candidate-base normalization

Put

$$\vartheta := \frac{\log 3}{\log 2} = \log_2 3 = 1.584962500721156 \dots$$

and define the *candidate set*

$$\mathcal{C} := \{m \in \mathbb{N}_{>0} : m^\vartheta \in \mathbb{N}\}.$$

Lemma 2.2 (Exponent–candidate bijection). *The maps $x \mapsto 2^x$ and $m \mapsto \log_2 m$ are mutually inverse bijections between $\mathcal{S} := \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{Z}\}$ and $\mathcal{C}$. Under the bijection, $3^x = m^\vartheta$.*

Proof. If $x \in \mathcal{S}$, set $m = 2^x \in \mathbb{N}$. Then $m^\vartheta = \exp(\log m \cdot \log 3 / \log 2) = \exp(x \log 3) = 3^x \in \mathbb{N}$. Conversely, if $m \in \mathcal{C}$ and $x = \log_2 m$, then $2^x = m$ and the same identity gives $3^x = m^\vartheta \in \mathbb{N}$. $\square$

Status: [\[kernel-verified Lean\]](#) —

`exists_twoBaseNaturalCandidate_of_two_three_rpow_integer,`
`TwoBaseNaturalCandidate.integer_powers.`

Integral solutions correspond exactly to powers of two (`twoBaseCandidateExponent_isInteger_iff_isPowerOfTwo`), so the conjecture is equivalent to

$$\mathcal{C} = \{2^n : n \in \mathbb{N}_0\}$$

(`alaogluErdosConjecture_iff_candidates_are_powers_of_two`). The candidate formulation is well suited to unique factorization, finite certification, finite differences of t^ϑ , and multiplicative-rank arguments, and it is the formulation in which most of the corpus is stated.

Two closure properties of $\mathcal{S}$ are elementary but essential: $\mathcal{S}$ is closed under addition and under multiplication by positive naturals, so it is an additive submonoid of $[0, \infty)$ containing $\mathbb{N}$ (`rpow_integer_add`, `rpow_integer_nat_mul`).

2.3 The four-exponentials reduction

Suppose x is nonintegral and $2^x, 3^x$ are algebraic. The rational-exponent argument (Lemma 2.4) shows that x cannot be rational, so $1, x$ are $\mathbb{Q}$ -linearly independent; and $\log 2, \log 3$ are $\mathbb{Q}$ -linearly independent since $2^a 3^b = 1$ forces $a = b = 0$. The four exponentials built from these two pairs are

$$e^{1 \cdot \log 2} = 2, \quad e^{1 \cdot \log 3} = 3, \quad e^{x \log 2} = 2^x, \quad e^{x \log 3} = 3^x,$$

all four algebraic — contrary to the four exponentials conjecture. This reduction is formalized: `alaogluErdosConjecture_of_realFourExponentialsConjecture`. [\[kernel-verified Lean\]](#)

The reduction is a diagnostic, not just context: any proposed elementary argument must exploit integrality, positivity, or arithmetic structure beyond bare algebraicity of those four values, or else it would prove a substantial case of the four exponentials conjecture by another name. By contrast, three multiplicatively independent bases fall to the six exponentials mechanism, and the special case needed here is fully formalized:

Theorem 2.3 (Three-base theorem). *If $2^x, 3^x, 5^x \in \mathbb{Z}$ for a real x , then $x \in \mathbb{Z}$.*

Status: [\[kernel-verified Lean\]](#) — modules `IntegerExponent`,

`IntegerExponent.DeterminantBound`, `IntegerExponent.ExponentialKernel`,
`IntegerExponent.Grid`, following Waldschmidt’s square-determinant proof [9].

In outline, for irrational x one forms a large square matrix $\mathcal{A}_{ij} = \exp(a_i b_j)$, where row arguments encode monomials in $2, 3, 5$ and column arguments encode affine forms in $1, x$. Irrationality makes the column parameters pairwise distinct, multiplicative independence the row parameters; a total-positivity argument gives $\det \mathcal{A} \neq 0$; integrality of all entries forces $|\det \mathcal{A}| \geq 1$; and an interpolation

estimate with explicit size choices gives $|\det \mathcal{A}| < 1$, a contradiction. The dimension count is critical: with three independent row directions and two column directions the analytic bound beats the arithmetic bound, while in the 2×2 case the same balance lands exactly on the classical boundary where decay below one fails. A successful two-base determinant would be a new four-exponentials argument.

The denominator-aware construction is symmetric enough to prove a rational form of the six-exponentials theorem without fixing the bases 2, 3. If $a, b, c \in \mathbb{Q}_{>0}$ have injective nonnegative monomials

$$(i, j, k) \mapsto a^i b^j c^k$$

and $a^x, b^x, c^x \in \mathbb{Q}$, then $x \in \mathbb{Q}$. Equivalently, for irrational x , three positive rational bases with rational x -th powers must satisfy a nontrivial multiplicative relation. This statement allows bases below 1 and negative exponents; its size estimate uses $1 + |x|$.

Status: [\[kernel-verified Lean\]](#) —

`rational_of_three_rat_rpows_rational_of_monomial_injective` and its contrapositive `not_monomial_injective_of_irrational_of_three_rat_rpows_rational` in `RationalSixExponentials`.

Denominator-unrestricted rational third outputs. The denominator restriction in the first version of the rational-output argument has now been removed. Let $a > 1$ and assume the monomial map

$$(i, j, k) \mapsto 2^i 3^j a^k$$

is injective. If $2^x, 3^x \in \mathbb{Z}$ and $a^x \in \mathbb{Q}$, then $x \in \mathbb{Q}$, and therefore $x \in \mathbb{Z}$. No divisibility hypothesis on the reduced denominator of a^x is needed. A prime divisor of a outside $\{2, 3\}$ supplies the required monomial injectivity.

The proof clears denominators inside the full interpolation grid. For side parameter n , each entry becomes integral after multiplication by $\text{den}(a^x)^{n^5}$; since the square matrix has size n^6 , the determinant scale is $\text{den}(a^x)^{n^{11}}$. More generally, if the third base itself is a positive rational q and $q^x = s \in \mathbb{Q}$, the common entry denominator is $(\text{den}(q) \text{den}(s))^{n^5}$. The explicit parameter choice in Lean absorbs the resulting n^{11} arithmetic lower bound into the analytic determinant margin. At the reusable level, a nonzero $m \times m$ real determinant whose entries become integers after multiplication by $Q > 0$ satisfies $|\det A| \geq Q^{-m}$; and for $m = 2r$, $r > 2$, $192C \leq r$, and $bs + 2r < r^2$, the verified analytic expression remains below 1 after multiplication by b^s .

Consequently, for any fixed hypothetical nonintegral solution,

$$a \in \mathbb{N}_{>0} : \quad a^x \in \mathbb{Q} \iff \exists u, v \in \mathbb{N}_0, \quad a = 2^u 3^v,$$

and the full positive-rational form is

$$q \in \mathbb{Q}_{>0} : \quad q^x \in \mathbb{Q} \iff \exists i, j, k, \ell \in \mathbb{N}_0, \quad q = \frac{2^i 3^j}{2^k 3^\ell}.$$

Status: [\[kernel-verified Lean\]](#) — `one_div_pow_le_abs_det_of_common_denominator`, `exponential_det_numeric_bound_with_denominator`, `rational_of_two_three_a_rpow_rational_of_monomial_injective`, and `rpow_rational_iff_eq_two_pow_mul_three_pow_of_not_integer` in `RationalThirdBase`; and `rational_of_two_three_rat_rpow_rational_of_monomial_injective`, `IsTwoThreeUnit`, and `rat_rpow_rational_iff_eq_two_three_ratio_of_not_integer` in `RationalBaseThird`. The earlier denominator-divisibility results remain available as specialized corollaries, but under monomial injectivity the determinant grid itself is now free of denominator-support hypotheses on rational third outputs.

Algebraic third outputs and the number-field norm. The same square determinant now handles an arbitrary algebraic value at a third natural base. Let $a > 1$, assume that

$$(i, j, k) \mapsto 2^i 3^j a^k$$

is injective, and suppose that $2^x, 3^x \in \mathbb{Z}$. If a^x is algebraic over $\mathbb{Q}$, then $x \in \mathbb{Q}$, hence $x \in \mathbb{Z}$. Thus the rational-output theorem above is not merely denominator-unrestricted: at an independent natural base even an algebraic irrational output is impossible.

The arithmetic lower bound is no longer obtained from a rational denominator alone. Place a^x in a number field K , choose a nonzero natural scale s for which $sa^x \in \mathcal{O}_K$, and form the interpolation matrix over $\mathcal{O}_K$. Each entry maps at the distinguished complex embedding to s^{n^5} times the corresponding analytic entry, so the determinant acquires the factor $s^{n^{11}}$. At every other embedding the six factor groups in an entry are bounded by T^{6n^5} . The field norm of the nonzero algebraic-integer determinant then gives

$$\frac{1}{((n^6)! T^{6n^{11}})^{[K:\mathbb{Q}]-1}} \leq |\sigma(\det A)|.$$

The verified estimates

$$(n^6)! \leq 2^{n^{11}}, \quad ((n^6)! T^{6n^{11}})^{[K:\mathbb{Q}]-1} \leq ((2T^6)^{[K:\mathbb{Q}]-1})^{n^{11}}$$

put the complete conjugate loss on the same n^{11} scale as the rational denominator loss. It is therefore absorbed by the existing explicit analytic margin.

Status: [\[kernel-verified Lean\]](#) — `one_div_conjugate_entry_bound_le_norm_det` and `one_div_entry_bound_le_scaled_norm_det` in `AlgebraicIntegerDeterminant`; `AlgebraicOutputBridge.ofRealIsAlgebraic` in `AlgebraicOutputBridge`; and `rational_of_two_three_a_rpow_isAlgebraic_of_monomial_injective` together with `integer_of_two_three_a_rpow_isAlgebraic_of_monomial_injective` in `AlgebraicThirdBase`.

At the fully symmetric level, let a, b, c be positive algebraic real numbers and suppose their nonnegative monomials are injective:

$$(i, j, k) \mapsto a^i b^j c^k.$$

If a^x, b^x, c^x are all algebraic, then $x \in \mathbb{Q}$. The theorem allows bases below one and negative x . Its six input/output values are placed in one finite number field with one common integral scale; the same n^{11} determinant bookkeeping absorbs the common denominator, every conjugate height, and the field degree. This is the positive-real algebraic six-exponentials interface underlying all of the specialized output classifications.

Status: [\[kernel-verified Lean\]](#) —
`rational_of_three_real_rpows_isAlgebraic_of_monomial_injective` in
`AlgebraicSixExponentials`, using
`FiniteAlgebraicOutputBridge.ofFinSixIsAlgebraic`.

The dependent case also has an exact arithmetic description. Define the positive 2, 3-radical hull by

$$\sqrt[n]{\langle 2, 3 \rangle_{\mathbb{Z}}} := \{\gamma > 0 : \exists n \geq 1, \gamma^n \in \langle 2, 3 \rangle_{\mathbb{Z}}\}.$$

For every positive algebraic real γ , under the two integrality hypotheses,

$$\text{Alg}_{\mathbb{Q}}(\gamma^x) \iff x \in \mathbb{Z} \quad \text{or} \quad \gamma \in \sqrt[n]{\langle 2, 3 \rangle_{\mathbb{Z}}}. \quad (1)$$

In particular, for a hypothetical nonintegral solution the algebraic-output locus among all positive algebraic bases is *exactly* this divisible hull. The arithmetic core is the equivalence between absence of a positive 2, 3-unit power and injectivity of $(i, j, k) \mapsto 2^i 3^j \gamma^k$; the forward implication in (1) then invokes the generic algebraic determinant, while the reverse implication descends algebraicity from a positive integral power.

Status: [\[kernel-verified Lean\]](#) — `HasPositiveTwoThreeUnitPower`,
`real_two_three_gamma_monomial_injective_iff`,
`TwoBaseNonintegerSolution.real_rpow_isAlgebraic_iff_hasPositiveTwoThreeUnitPower`,
and `real_rpow_isAlgebraic_iff_integer_or_hasPositiveTwoThreeUnitPower` in
`AlgebraicBaseUnitPower`.

There is a coordinate-free rank version. Let the algebraic output locus have rank three when there exist positive algebraic a, b, c with injective nonnegative monomials and with a^x, b^x, c^x all algebraic. Under $2^x, 3^x \in \mathbb{Z}$, this rank-three property holds if and only if $x \in \mathbb{Z}$: the difficult implication is the generic determinant theorem, while the reverse implication uses 2, 3, 5. Universalizing gives another exact equivalence with AE.

Status: [\[kernel-verified Lean\]](#) — `AlgebraicRpowOutputLocusHasRankThree`,
`algebraicRpowOutputLocusHasRankThree_iff_integer`, and
`alaogluErdosConjecture_iff_algebraicRpowOutputLocusHasRankThree` in
`AlgebraicRankThreeEquivalence`.

Combining that determinant theorem with the elementary factorization of natural numbers gives the exact unconditional pointwise classification

$$\text{Alg}_{\mathbb{Q}}(a^x) \iff x \in \mathbb{Z} \quad \text{or} \quad a = 2^u 3^v \text{ for some } u, v \in \mathbb{N}_0, \quad (2)$$

for every $a \in \mathbb{N}_{>0}$ whenever $2^x, 3^x \in \mathbb{Z}$. Thus on a hypothetical nonintegral solution, algebraic, rational, and integral outputs at a positive natural base are all equivalent, and occur precisely for 2,3-smooth bases. The primitive odd and 3-free cores consequently have algebraic x -th powers exactly in the exceptional cases 3 and 2, respectively; every other such core has transcendental output.

For a positive rational base the exact statement is equally sharp:

$$\text{Alg}_{\mathbb{Q}}(q^x) \iff x \in \mathbb{Z} \text{ or } q \in \langle 2, 3 \rangle_{\mathbb{Z}} \quad (q \in \mathbb{Q}_{>0}),$$

where $\langle 2, 3 \rangle_{\mathbb{Z}} = \{2^i 3^j / (2^k 3^\ell) : i, j, k, \ell \in \mathbb{N}_0\}$. Thus for a hypothetical nonintegral solution, algebraicity and rationality of q^x are equivalent and characterize exactly the positive 2,3-units.

Status: [\[kernel-verified Lean\]](#) —

`rat_rpow_isAlgebraic_iff_isTwoThreeUnit_of_not_integer,`
`twoBaseNonintegerSolution_rat_rpow_isAlgebraic_iff_isTwoThreeUnit,` and
`rat_rpow_isAlgebraic_iff_integer_or_isTwoThreeUnit` in `AlgebraicRationalBase`.

The classification also constrains iteration. A hypothetical nonintegral solution obeys

$$\text{Trans}_{\mathbb{Q}}(2^{x^2}) \text{ or } \text{Trans}_{\mathbb{Q}}(3^{x^2}), \quad \text{Trans}_{\mathbb{Q}}((2^i 3^j)^{x^2}) \quad (i, j > 0).$$

Indeed, write $M = 2^x$, $B = 3^x$. One of the two complete monomial half-planes is transcendental: either every $(M^r B^s)^x$ with $r > 0$ is transcendental, or every such output with $s > 0$ is. If a positive mixed monomial were 2,3-smooth, both positive divisors M and B would be smooth, forcing both first iterates to be algebraic, contrary to the simultaneous-core obstruction.

There is also a rank-one restriction on rational second-iterate ratios. If $il \neq kj$, the two numbers

$$\left(\frac{2^i}{3^j}\right)^{x^2}, \quad \left(\frac{2^k}{3^\ell}\right)^{x^2}$$

cannot both be algebraic. Thus all algebraic members of this two-parameter family lie on one rational ray in the exponent lattice. More precisely, a counterexample supplies positive outputs $M = 2^x$, $B = 3^x$ and a prime $p \notin \{2, 3\}$ dividing M or B such that every algebraic ratio output satisfies the single nontrivial valuation equation

$$i v_p(M) = j v_p(B).$$

This gives one explicit p -adic line containing the entire algebraic locus.

Three concise equivalent formulations of the original conjecture result:

$$\begin{aligned} \text{AE} &\iff \forall x, 2^x, 3^x \in \mathbb{Z} \implies \text{Alg}_{\mathbb{Q}}(5^x), \\ &\iff \forall x, 2^x, 3^x \in \mathbb{Z} \implies \text{Alg}_{\mathbb{Q}}((2^x)^x) \wedge \text{Alg}_{\mathbb{Q}}((3^x)^x), \\ &\iff \forall x, 2^x, 3^x \in \mathbb{Z} \implies \text{Alg}_{\mathbb{Q}}(6^{x^2}). \end{aligned}$$

There is also an intrinsic rank formulation with no fixed auxiliary output. Say that the rational squared-output locus has rank two when it contains two algebraic members $(2^i/3^j)^{x^2}$ and $(2^k/3^\ell)^{x^2}$ with $il \neq kj$. Under the two integrality hypotheses this happens if and only if $x \in \mathbb{Z}$. Hence AE is equivalent to requiring this rank-two property for every solution. A counterexample would force the entire algebraic locus onto a single rational ray.

Status: [\[kernel-verified Lean\]](#) —

`rpow_isAlgebraic_iff_integer_or_eq_two_pow_mul_three_pow`,
`TwoBaseNonintegerSolution.rpow_isAlgebraic_iff_rational`,
`TwoBaseNonintegerSolution.transcendental_squared_exponent_output`,
`TwoBaseNonintegerSolution.transcendental_six_squared_exponent_output`, and the
three equivalences `alaogluErdosConjecture_iff_five_rpow_isAlgebraic`,
`alaogluErdosConjecture_iff_both_iterated_outputs_isAlgebraic`, and
`alaogluErdosConjecture_iff_six_squared_exponent_isAlgebraic` in
`AlgebraicConsequences`; the half-plane, all mixed powers, and rank-one ratio locus are
`exists_transcendental_output_monomial_halfplane`,
`transcendental_mixed_two_three_squared_exponents`, and
`not_both_algebraic_two_three_ratio_squared_of_det_ne_zero` in
`AlgebraicOutputLocus`; the pointwise and global rank-two equivalences are
`ratioSquaredAlgebraicLocusHasRankTwo_iff_integer` and
`alaogluErdosConjecture_iff_ratioSquaredAlgebraicLocusHasRankTwo`, while the
explicit valuation line is `exists_external_prime_algebraic_output_ratio_padic_line`.

2.4 Rational and algebraic exponents

Lemma 2.4 (Rational exponent). *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$. Consequently every nonintegral solution is irrational.*

Proof. Write $x = a/b$ in lowest terms with $b > 0$. If $2^{a/b} = m \in \mathbb{N}$, then $2^a = m^b$; comparing the exponent of 2 gives $a = b v_2(m)$, so $b \mid a$ and $b = 1$. $\square$

Status: [\[kernel-verified Lean\]](#) —

`IntegerExponent.integer_of_rational_of_two_rpow_integer`,
`irrational_of_not_integer_of_two_rpow_integer`.

Gelfond–Schneider upgrades irrational to transcendental; see [section 4.6](#).

3 Repository timeline and verification status

The formal development advanced in three waves plus documentation. All commits are in the repository [\[13\]](#); the mathematical role of each is summarized below.

Commit	Role
f492c18f1	Three-base theorem via interpolation determinants (<code>IntegerExponent</code> and submodules).
5ed4e6185	Merge of the three-base branch.

Commit	Role
a37d97f9d	Basic two-base module: the conjecture as a proposition, closure under natural multiples, irrationality of nonintegral solutions, endpoint gaps, numerical bounds for ϑ , the $2^x < 10$ exclusion, second-difference obstruction, aligned-block 2/3 counting.
4a9b03d12	Major corpus: finite checks through $2^x < 256$; arithmetic descent; multiplicative rank; candidate structure; zero density; shrinking targets; the 3-smooth theorem; the four-exponentials reduction; transcendence consequences; and a complete Gelfond–Schneider port.
18a85e6b8	Merge of the two-base branch.
423ac24f0	First survey document (<code>two-base-partial-results.tex</code>) with rendered PDF.
98646b455	Three further modules, kernel-verified: <code>FractionalPartDensity</code> (density dichotomy and gap equivalences), <code>OddCore</code> (common odd core, improved counting, dyadic blocks), <code>FiniteCheck4096</code> (zero-free region to $2^x < 4096$).
716617d72	Survey finalized after kernel acceptance of the 4096 region.
11ac3093f	Independent research report (audit at 423ac24f0, exact conditional structure, interval scan to 2^{20}); one of the two sources merged into the present document.
<i>current tree</i>	Kernel-checked feature modules formalizing the primitive common odd core, least-solution primitivity, simultaneous output normalization, power-of-three denominator normalization, rational-power index and exact primitive generator, exact conditional counting and its quadratic asymptotic, finite-Fourier block averages, a generic rational six-exponentials theorem, denominator-unrestricted natural and rational third-base outputs under monomial injectivity, canonical-core rationality restrictions, primitive localized-radical reduction, exact radical degree and normalized-radical output restrictions, and uniform higher finite differences.

The audit performed in 11ac3093f treated results of the then-uncommitted modules as article-level claims; with 98646b455 in the tree and kernel-verified, those results are now cited as committed Lean. The interval scan to 2^{20} remains at the [\[interval computation\]](#) level by design. The later feature modules listed in the final row prove the cumulative asymptotic and finite-trigonometric-polynomial block averages. They do not yet formalize the stronger continuous-test-function or interval-counting equidistribution theorem; that distinction is maintained throughout this report.

4 The kernel-verified corpus

Throughout this section x denotes a hypothetical nonintegral solution, $m = 2^x \in \mathbb{N}$, $n = 3^x \in \mathbb{N}$. Every statement here is [\[kernel-verified Lean\]](#) unless labeled otherwise.

4.1 Fractional-part gaps and shrinking targets

An integral power caught strictly between consecutive integral powers of its base must clear both by at least one full unit; dividing through normalizes this to the fractional part.

Theorem 4.1 (Fractional-part gaps). *Let $x \in (N, N+1)$ with $N \in \mathbb{N}$, $t = x - N$, and suppose $2^x, 3^x \in \mathbb{Z}$. Then*

$$1 + 2^{-N} \leq 2^t \leq 2 - 2^{-N}, \quad 1 + 3^{-N} \leq 3^t \leq 3 - 3^{-N}.$$

Status: [\[kernel-verified Lean\]](#) — `two_three_fractional_part_gaps`,
`two_three_rpow_gaps_near_nat`, from
`rpow_integer_gap_between_consecutive_powers`.

Near the endpoints the excluded zones have width $\asymp 2^{-N}$: they *shrink* as the integer part grows. Applied to all multiples kx (which are again solutions), this yields explicit Diophantine lower bounds.

Theorem 4.2 (Effective irrationality bound). *Let $x \notin \mathbb{Z}$ satisfy $2^x \in \mathbb{Z}$ and write $m = 2^x$. Then for all integers p and all $k \geq 1$,*

$$\left| x - \frac{p}{k} \right| \geq \frac{1}{2km^k}.$$

Exact (slightly stronger) radii, and two-base versions taking the better of the base-2 and base-3 radii, are formalized as well.

Status: [\[kernel-verified Lean\]](#) — `rational_approximation_exponential_lower_bound`,
`two_three_multiple_distance_to_integer_lower_bound`,
`exists_natural_base_rational_approximation_lower_bound` in `ShrinkingTarget`.

In shorthand, $\|kx\|_{\mathbb{Z}} \gtrsim 2^{-kx}$. This bound is compatible with ordinary Diophantine approximation: Dirichlet guarantees infinitely many k with $\|kx\|_{\mathbb{Z}} < 1/k$, and the shrinking radius 2^{-kx} is incomparably smaller. The interplay between this exponential gap and equidistribution is discussed in [section 9](#).

The bound dualizes into a growth constraint on the continued fraction of a solution: if $|x - p/q| < 1/(qQ)$ with $q, Q \geq 1$, then $Q < 2m^q$. Applied to consecutive convergent denominators, which satisfy $|x - p_n/q_n| < 1/(q_n q_{n+1})$, this gives

$$q_{n+1} < 2m^{q_n} :$$

the partial quotients of a hypothetical nonintegral solution are capped by a single exponential in the preceding denominator, with base its own output $m = 2^x$.

Status: [\[kernel-verified Lean\]](#) —
`approximation_quality_lt_of_noninteger_solution` in `DenominatorGrowth`.

4.2 Verified zero-free regions and certificate architecture

Theorem 4.3 (Zero-free region). *If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 4096$, then $x \in \mathbb{Z}$. Equivalently, every nonintegral solution satisfies $x \geq 12$, and every candidate that is not a power of two is at least 4096.*

Status: [\[kernel-verified Lean\]](#) —

`integer_of_two_three_rpow_integer_of_two_rpow_lt_4096,`
`twelve_le_of_not_integer_of_two_three_rpow_integer,`
`le_of_twoBaseNonintegerCandidate_4096` in `FiniteCheck4096`, extending `FiniteCheck`
 $(2^x < 100)$ and `FiniteCheck256` ($2^x < 256$, hence $x \geq 8$).

The certificate architecture is worth recording, both because it is the repository’s model of a trustworthy large computation and because scaling it is a named research direction. For each non-power-of-two m the certificate is a pair of exact integer power comparisons against rational enclosures $L_p/L_q < \vartheta < U_p/U_q$ drawn from continued-fraction convergents of ϑ : from

$$a^{L_q} < m^{L_p} \quad \text{and} \quad m^{U_p} < (a+1)^{U_q}$$

one traps $a < m^\vartheta < a+1$, so $m^\vartheta \notin \mathbb{Z}$. The $2^x < 256$ stage verifies per-value certificates by `norm_num` with three enclosure tiers (exponent sizes ~ 500 , ~ 1500 , ~ 25000). The extension to 4096 verifies 3836 certificates through a single kernel-replayed `decide` over a table generated by PARI/GP and revalidated independently with exact big-integer arithmetic in Python. Five tiers suffice, with tier populations 78, 488, 3241, 28, 1:

tier	lower convergent L_p/L_q	upper convergent U_p/U_q
0	569/359	485/306
1	1054/665	1539/971
2	25781/16266	24727/15601
3	50508/31867	125743/79335
4	176251/111202	301994/190537

Only the value $m = 2762$, whose ϑ -power is within $3 \cdot 10^{-9}$ of an integer in logarithmic scale, needs tier 4; its certificate compares integers of roughly 90,000 digits, which the kernel’s exact arithmetic replays without difficulty. No axioms beyond the Lean kernel are involved; in particular no `native_decide`.

4.3 Arithmetic descent

Theorem 4.4 (Affine descent). *If the two integral outputs factor simultaneously as $2^x = 2^r u^k$ and $3^x = 3^r v^k$ with $r \geq 0$, $k \geq 1$, then $(x-r)/k$ is again a solution. Combined with the $x \geq 8$ region this gives the unconditional constraint*

$$r + 8k \leq x$$

for nonintegral x ; in particular common perfect-power shapes of the two outputs are strongly restricted.

Status: [\[kernel-verified Lean\]](#) — `affine_descent_integral_powers`, `base_depth_add_eight_mul_degree_le`, `min_output_base_factorization_add_eight_le` in `ArithmeticRigidity`. With the 4096 region the constant 8 improves to 12 by bookkeeping; with the interval scan of [section 8](#) it would improve to 20 at the [\[interval computation\]](#) level.

For the *least* counterexample the same descent later becomes an exact primitivity statement (Corollary 5.6): no nontrivial matched decomposition exists at all.

4.4 Local progression obstructions and zero density

The function $t \mapsto t^\vartheta$ is strictly convex with second differences at moderate steps lying strictly between 0 and 1 — hence never integers.

Theorem 4.5 (Committed AP obstruction). *Let $n, h \geq 1$ with $h^5 \leq n$. Then $n, n + h, n + 2h$ are not all candidates. In particular no three consecutive naturals are candidates, and among the positive integers up to $3N$ at most $2N$ are candidates.*

Status: [\[kernel-verified Lean\]](#) — `not_three_arithmetic_progression_twoBaseNaturalCandidates`, `not_three_consecutive_logThreeDivLogTwo_rpow_integers`, `twoBaseNaturalCandidate_lcc_card_le`. The sharper criterion $\vartheta(\vartheta - 1)h^2 < n^{2-\vartheta}$ is kernel-verified in `SharperProgressions`, and the analogous four-term criterion is kernel-verified in `ThirdDifference`. The arbitrary-order statement for every $r \geq 2$ is `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` in `HigherDifference`.

The hypothesis bounding h is essential: for steps comparable to n the second difference is large and the obstruction vanishes. This locality is a genuine barrier — a *global* progression obstruction would, for example, forbid candidates with odd part $2^s \pm 1$ via progressions such as $(2^i, 2^{i+t-1}, 2^i(2^t - 1))$, but no such global statement is available; under a counterexample the exact monoid of [section 5](#) shows sparse candidates whose additive relations the curvature argument cannot see.

Feeding the local obstruction into the theory of Roth numbers gives quantitative sparsity.

Theorem 4.6 (Zero density). *For every $H \geq 1$ the number of candidates below N is at most $H^5 + \lceil (N - H^5)/H \rceil r_3(H)$, where $r_3(H)$ is the maximal size of a 3-AP-free subset of $\{0, \dots, H - 1\}$. Consequently the candidate counting function is $o(N)$: the candidates, and a fortiori the values 2^x of nonintegral solutions, have asymptotic density zero.*

Status: [\[kernel-verified Lean\]](#) — `twoBaseNaturalCandidatesBelow_card_le`, `twoBaseNaturalCandidatesBelow_isLittleO_id`, `tendsto_twoBaseNonintegerCandidatesBelow_density_zero` in `ZeroDensity` and `NonintegerDensity`.

4.5 Multiplicative rank two and the common odd core

Theorem 4.7 (Rank two). *The prime-exponent vectors $\mathbf{v}(a) = (v_p(a))_p$ of all candidates span a rational space of dimension at most two. Consequently at most $(\log_2 N)^2$ candidates lie below N .*

Status: [\[kernel-verified Lean\]](#) — `finrank_span_primeExponentVectors_le_two`, `twoBaseNaturalCandidatesBelow_card_le_clog_sq` in `MultiplicativeRank`. The mechanism abstracts the determinant method: three candidates with independent exponent vectors would give three multiplicatively independent bases, contradicting Theorem 2.3 through the irrationality of ϑ (`irrational_logThreeDivLogTwo`).

Since 2 is itself a candidate, all odd parts lie on at most one further rational direction. Pairwise this yields the cross-valuation identity, and globally a common odd core.

Lemma 4.8 (Cross-valuation identity). *Let $a, b \in \mathcal{C}$ and let p be an odd prime with $v_p(a) > 0$. Then for every odd prime q ,*

$$v_p(a) v_q(b) = v_p(b) v_q(a).$$

Status: [\[kernel-verified Lean\]](#) — `odd_factorization_cross_mul` in `CandidateStructure`; the fixed-power relations `fixed_power_relation_of_odd_prime_factor` follow.

Theorem 4.9 (Common odd core). *There is a single odd $w > 1$ such that every candidate equals $2^i w^j$ for some $i, j \in \mathbb{N}_0$; candidates that are not powers of two have $j \geq 1$. In exponent coordinates: every solution has the form $x = i + j \log_2 w$ with $i, j \in \mathbb{N}_0$.*

Status: [\[kernel-verified Lean\]](#) — `exists_common_odd_core`, `exists_common_odd_core_split`, `exists_common_odd_core_exponent` in `OddCore`. The canonical (gcd-normalized) version with uniqueness of w and of each coordinate pair is `existsUnique_primitive_common_odd_core` in `PrimitiveOddCore`.

Corollary 4.10 (Improved counting). *The number of candidates below N is at most $\log_2 N \cdot \log_3 N$, and each dyadic block $[2^T, 2^{T+1})$ contains at most $\log_3(2^{T+1}) \approx 0.631(T+1)$ candidates. In exponent coordinates, the number of solutions with $x \in [T, T+1)$ grows at most linearly in T , against the quadratic global bound.*

Status: [\[kernel-verified Lean\]](#) — `twoBaseNaturalCandidatesBelow_card_le_clog_mul_clog`, `twoBaseNonintegerCandidatesBelow_card_le_clog_mul_clog`, `twoBaseNaturalCandidatesInDyadicBlock_card_le` in `OddCore`.

The logarithmic-square order cannot be improved by counting alone: if a candidate with an odd prime factor exists, multiplicative closure generates injective boxes $2^i a^j$, so the count below $2^k a^l$ is at least kl (`twoBaseNaturalCandidatesBelow_card_ge_box`, `twoBaseNonintegerCandidatesBelow_card_ge_box`) — a conditional lower bound of the same quadratic-in-logarithm shape, made exact in [section 5.4](#).

4.6 Transcendence

The corpus contains a complete Lean proof of the Gelfond–Schneider theorem (Hilbert’s seventh problem), following Hua’s exposition [4], ported from Michail Karatarakis’s Mathlib formalization (Mathlib PR #42911, Apache 2.0).

Theorem 4.11 (Gelfond–Schneider). *If α, β are algebraic, $\alpha \neq 0, 1$, and β is irrational, then α^β is transcendental.*

Status: [\[kernel-verified Lean\]](#) —

`GelfondSchneider.transcendental_cpow_of_isAlgebraic_of_irrational.`

Corollary 4.12 (Algebraic–integral dichotomy). *If $2^x \in \mathbb{Z}$ then x is either an integer or transcendental; algebraicity of such x is equivalent to integrality. Every nonintegral two-base solution is transcendental, and $\vartheta = \log_2 3$ is transcendental.*

Status: [\[kernel-verified Lean\]](#) —

`transcendental_of_not_integer_of_two_rpow_integer,`
`isAlgebraic_iff_integer_of_two_rpow_integer,`
`transcendental_of_not_integer_of_two_three_rpow_integer,`
`transcendental_logThreeDivLogTwo.`

The dichotomy does not resolve the conjecture — the unknown exponent is allowed to be transcendental — but it constrains every construction below: the primitive generator β of [section 5](#) is transcendental, as is $\log_2 w$ for the odd core.

4.7 Classification of auxiliary natural and rational bases

Theorem 4.13 (3-smooth classification). *Suppose $x \notin \mathbb{Z}$ and $2^x, 3^x \in \mathbb{Z}$. Then for a positive natural a , the power a^x is an integer if and only if $a = 2^u 3^v$ for some $u, v \in \mathbb{N}_0$. In particular a third base with any prime factor other than 2 and 3 forces $x \in \mathbb{Z}$.*

Status: [\[kernel-verified Lean\]](#) —

`rpow_integer_iff_eq_two_pow_mul_three_pow_of_not_integer,`
`integer_of_two_three_a_rpow_integer_of_prime_factor in ThreeSmooth.`

The denominator-unrestricted rational-output determinant sharpens “integer” to “rational” without changing the natural bases that survive, and also classifies all positive rational bases.

Theorem 4.14 (Rational-output classification). *Suppose $x \notin \mathbb{Z}$ and $2^x, 3^x \in \mathbb{Z}$. Then*

$$a^x \in \mathbb{Q} \iff \exists u, v \in \mathbb{N}_0, a = 2^u 3^v \quad (a \in \mathbb{N}_{>0}),$$

and

$$q^x \in \mathbb{Q} \iff \exists i, j, k, \ell \in \mathbb{N}_0, q = 2^i 3^j / (2^k 3^\ell) \quad (q \in \mathbb{Q}_{>0}).$$

Status: [\[kernel-verified Lean\]](#) —

`rpow_rational_iff_eq_two_pow_mul_three_pow_of_not_integer` in
`RationalThirdBase`, and `rat_rpow_rational_iff_eq_two_three_ratio_of_not_integer`
in `RationalBaseThird`.

These are exact classifications of bases for a *fixed* hypothetical counterexample. They should not be confused with the classification of candidate values 2^x as x ranges over all solutions, which is the subject of the next section. Note also the scope limitation explained in [section 9](#): the theorem controls the quantity b^x , not divisibility of the integer 2^x by b .

4.8 Unconditional gap equivalences

Because natural multiples of a solution are solutions, one nonintegral solution generates infinitely many, with irrational generator; Dirichlet approximation and a rotation-stepping argument spread their fractional parts densely.

Theorem 4.15 (Density dichotomy). *If some $x \notin \mathbb{Z}$ satisfies $2^x, 3^x \in \mathbb{Z}$, then for every $t \in [0, 1]$ and $\varepsilon > 0$ there is a nonintegral solution y with $|\{y\} - t| < \varepsilon$; and the set of nonintegral solutions is infinite.*

Theorem 4.16 (Gap equivalences). *Each of the following is equivalent to Conjecture 2.1:*

- (i) *some nonempty open interval $(u, v) \subseteq (0, 1)$ contains no fractional part of a nonintegral solution;*
- (ii) *the fractional parts of nonintegral solutions are bounded away from 1;*
- (iii) *the fractional parts of nonintegral solutions are bounded away from 0.*

Status: [\[kernel-verified Lean\]](#) — `dense_fract_of_noninteger_solution`,
`infinite_noninteger_solutions_of_exists`,
`alaogluErdosConjecture_iff_fract_avoids_interval`,
`alaogluErdosConjecture_iff_fract_bounded_away_one`,
`alaogluErdosConjecture_iff_fract_bounded_away_zero` in `FractionalPartDensity`,
built on the reusable `exists_pos_nat_fract_close_of_irrational`.

Remark 4.17 (Consistency pincer). Theorem 4.1 bounds $\{x\}$ away from 0 and 1 by margins $\asymp 2^{-\lfloor x \rfloor}$ that decay with the height of the solution, while Theorem 4.15 produces fractional parts arbitrarily close to 0 and 1 along multiples of unbounded height. The two results meet edge-to-edge without contradiction, and Theorem 4.16 shows the decay is essential: *any* height-independent gap, however small and wherever located, is exactly as strong as the full conjecture. The elementary gap bounds are therefore of optimal shape, and the fractional-part axis measures the entire remaining distance to Conjecture 2.1.

These equivalences are *unconditional* biconditionals. Conditional on failure, the exact structure of the next section gives kernel-verified convergence for every finite trigonometric polynomial and the paper-level strengthening to full blockwise Weyl equidistribution (Theorem 5.11).

5 Exact conditional structure of all solutions

This section develops the exact rigidity theory of a hypothetical counterexample. The canonical primitive odd core, rational-power index, denominator normalization, unique least solution, simultaneous primitive output normalization, exact free-monoid classification, exact unit-block counts, the cumulative quadratic asymptotic, and finite-Fourier block averages are now [\[kernel-verified Lean\]](#). Full blockwise Weyl equidistribution for arbitrary continuous tests or interval indicators remains [\[paper proof in this report\]](#).

5.1 The primitive common odd core

Theorem 5.1 (Primitive common odd core). *Assume $\mathcal{C}$ contains a non-power of two. Then there is a unique odd integer $w > 1$ that is power-free in the sense*

$$\gcd\{v_p(w) : p \mid w\} = 1,$$

and every candidate $b \in \mathcal{C}$ has a unique representation

$$b = 2^i w^j, \quad i, j \in \mathbb{N}_0.$$

Proof. Choose a non-power-of-two candidate a . Its odd valuation vector $A = (A_q)_{q \neq 2}$, $A_q = v_q(a)$, is nonzero and finitely supported. Let

$$g = \gcd\{A_q : A_q > 0\}, \quad c_q = A_q/g, \quad w = \prod_{q \neq 2} q^{c_q}.$$

Then w is odd, greater than one, and the positive c_q have gcd one.

Let $b \in \mathcal{C}$ and set $B_q = v_q(b)$. Fix an odd prime p with $A_p > 0$. By Lemma 4.8, $A_p B_q = B_p A_q$ for all odd q . If $A_q = 0$, this forces $B_q = 0$. On the support of A it says

$$B_q = \frac{B_p}{A_p} A_q = \left(\frac{g B_p}{A_p} \right) c_q.$$

Put $t = g B_p / A_p \in \mathbb{Q}_{\geq 0}$. Every $t c_q$ is an integer. Since the c_q have gcd one, there are integers r_q , finitely many nonzero, with $\sum r_q c_q = 1$; therefore $t = \sum r_q (t c_q) \in \mathbb{Z}$. Writing $j = t \in \mathbb{N}_0$, the odd part of b is exactly w^j , so $b = 2^i w^j$ with $i = v_2(b)$.

Uniqueness of (i, j) for fixed w follows from odd valuations and the 2-adic valuation. If w' is another primitive common core, the odd valuation vectors of w and w' generate the same rational ray and are both primitive integer vectors, hence equal. $\square$

Status: [\[kernel-verified Lean\]](#) — `existsUnique_primitive_common_odd_core` in `PrimitiveOddCore`. The module defines `oddPrimeValuationGCD`, proves its equivalence with power-theoretic primitivity for $w > 1$, and kernel-checks both uniqueness of the normalized core and uniqueness of every pair (i, j) .

The gcd normalization is what makes w canonical: it prevents hidden perfect powers from being absorbed into the exponent. Thus the earlier existence theorem for an arbitrary common core and the normalization argument above now both have kernel-verified counterparts.

Set

$$\alpha := \log_2 w, \quad E := 3^\alpha = w^\vartheta.$$

Every candidate exponent then has the form $x = i + j\alpha$. The number α is irrational (else the rational-exponent lemma applied to $2^\alpha = w$ would make it integral, forcing the odd $w > 1$ to be a power of two), and indeed transcendental by Corollary 4.12. The candidate condition becomes

$$3^{i+j\alpha} = 3^i E^j \in \mathbb{N},$$

and at this stage the admissible pairs (i, j) still form an unknown subset of $\mathbb{N}_0^2$. The next subsection shows this subset is completely rigid.

5.2 The rational-power index and the least generator

Because a non-power-of-two candidate exists, some $j > 0$ has $E^j \in \mathbb{Q}$. Let

$$d := \min\{j \in \mathbb{N}_{>0} : E^j \in \mathbb{Q}\}.$$

Lemma 5.2 (All rational powers are multiples of the least one). *For every $j \in \mathbb{Z}$: $E^j \in \mathbb{Q} \iff d \mid j$.*

Proof. $J := \{j \in \mathbb{Z} : E^j \in \mathbb{Q}\}$ is an additive subgroup of $\mathbb{Z}$ ($0 \in J$; products add exponents; reciprocals negate them), and every nonzero subgroup of $\mathbb{Z}$ is $d\mathbb{Z}$ for its least positive element. $\square$

Status: [\[kernel-verified Lean\]](#) — `rationalPowerIndices`, `AddSubgroup.Int.exists_least_positive_generator`, and `exists_rationalPowerIndex` in `RationalPowerIndex`.

Write the positive rational E^d in lowest terms. Its denominator has no prime other than 3: taking a candidate with odd-core exponent $j = dk > 0$ and 2-exponent i , integrality of $3^i(E^d)^k$ shows any other prime in the reduced denominator would survive in the k -th power. Hence there are unique integers $a \geq 0$, $A \geq 1$ with

$$E^d = \frac{A}{3^a}, \tag{3}$$

and $3 \nmid A$ when $a > 0$. Define

$$\beta := a + d\alpha = a + d \log_2 w, \quad M := 2^\beta = 2^a w^d, \quad A = 3^\beta = 3^a E^d. \tag{4}$$

Thus β is itself a solution, irrational because α is.

Status: [\[kernel-verified Lean\]](#) — the denominator-support step is `exists_three_pow_denominator_of_rational_power` in `ThreeDenominatorNormalization`; the normalized integrality criterion is `three_pow_mul_normalized_pow_integer_iff` in `RationalPowerIndex`.

Theorem 5.3 (Exact primitive-generator theorem). *Assume Conjecture 2.1 is false. With w, d, a, β, M, A as above:*

- (i) every solution has a unique representation $x = n + k\beta$ with $n, k \in \mathbb{N}_0$;
- (ii) every candidate has a unique representation $m = 2^n M^k$ with $n, k \in \mathbb{N}_0$;
- (iii) every output pair has a unique representation $(2^x, 3^x) = (2^n M^k, 3^n A^k)$;
- (iv) β is the unique least nonintegral solution.

Proof. Let $x \in \mathcal{S}$. By Theorem 5.1, $2^x = 2^i w^j$ for unique $i, j \in \mathbb{N}_0$, so $x = i + j\alpha$. From $3^x = 3^i E^j \in \mathbb{N}$ we get $E^j \in \mathbb{Q}$, so Lemma 5.2 gives $j = dk$. Using (3),

$$3^x = 3^i (E^d)^k = 3^{i-ak} A^k.$$

Because $A/3^a$ is reduced, this is an integer exactly when $i \geq ak$. Put $n = i - ak \in \mathbb{N}_0$; then

$$x = i + dk\alpha = n + k(a + d\alpha) = n + k\beta.$$

Conversely, for any $n, k \in \mathbb{N}_0$,

$$2^{n+k\beta} = 2^n M^k \in \mathbb{N}, \quad 3^{n+k\beta} = 3^n A^k \in \mathbb{N},$$

so $n + k\beta \in \mathcal{S}$. If $n + k\beta = n' + k'\beta$ with $k \neq k'$, then $\beta = (n' - n)/(k - k') \in \mathbb{Q}$, a contradiction; hence the representation is unique. This proves (i), and (ii)–(iii) follow by exponentiation. A nonintegral solution has $k \geq 1$; the least value is attained at $(n, k) = (0, 1)$ and equals β , uniquely so by uniqueness of representation. $\square$

Status: [\[kernel-verified Lean\]](#) —

`exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture` in `PrimitiveGenerator`, built from

`exists_exact_solution_monoid_of_fixed_common_oddCore` in `RationalPowerIndex`.

Conditional on failure, $\mathcal{S}$ is not merely contained in a rank-two additive set: every solution has a unique pair of coordinates in the free additive monoid on 1 and the irrational least generator β . The set of occurring odd-core exponents is exactly $d\mathbb{N}_0$, with the power of two governed by one linear threshold.

Packaged algebraically: $(\mathbb{N}_0^2, +) \xrightarrow{\sim} (\mathcal{S}, +)$ via $(n, k) \mapsto n + k\beta$, and $(\mathbb{N}_0^2, +) \xrightarrow{\sim} (\mathcal{C}, \cdot)$ via $(n, k) \mapsto 2^n M^k$; the output-pair monoid $\mathcal{P}$ is freely generated by $(2, 3)$ and (M, A) . Every multiplicative relation among candidates is the obvious coordinate relation. This exact conditional model is the benchmark against which any future density, progression, or descent argument should be tested.

5.3 Primitivity of the least output pair

Leastness of β upgrades the repository’s descent inequalities to exact exclusions.

Corollary 5.4 (No matched base depth). *For the primitive pair $(M, A) = (2^\beta, 3^\beta)$: $\min\{v_2(M), v_3(A)\} = 0$; that is, M is odd or $3 \nmid A$ (possibly both).*

Proof. If both valuations were positive, then $2^{\beta-1} = M/2$ and $3^{\beta-1} = A/3$ would be integers, so $\beta - 1$ would be a smaller nonintegral solution. $\square$

Corollary 5.5 (No common perfect-power degree). *There is no $k \geq 2$ for which both M and A are perfect k -th powers.*

Proof. If $M = U^k$, $A = V^k$, then $2^{\beta/k} = U$ and $3^{\beta/k} = V$, and β/k is a smaller nonintegral solution. $\square$

Corollary 5.6 (Exact affine primitivity). *If $M = 2^r U^k$ and $A = 3^r V^k$ with $r \in \mathbb{N}_0$, $k \in \mathbb{N}_{>0}$, $U, V \in \mathbb{N}$, then $r = 0$ and $k = 1$.*

Proof. Affine descent (Theorem 4.4) gives the solution $(\beta - r)/k$, nonintegral since β is irrational; if $r > 0$ or $k > 1$ it is strictly smaller than β . $\square$

Status: [\[kernel-verified Lean\]](#) — `exists_least_twoBaseNonintegerSolution`, `IsLeastTwoBaseNonintegerSolution.unique`, `IsLeastTwoBaseNonintegerSolution.not_both_base_dvd_outputs`, `IsLeastTwoBaseNonintegerSolution.no_common_perfect_power`, and `IsLeastTwoBaseNonintegerSolution.affine_primitive` in `LeastSolution`.

Theorem 5.7 (Simultaneous primitive output normalization). *If the conjecture fails, its least nonintegral solution β admits simultaneous factorizations*

$$2^\beta = 2^a w^d, \quad 3^\beta = 3^b v^e,$$

where $w > 1$ is odd and power-primitive, $v > 1$ is power-primitive with $3 \nmid v$, and $d, e > 0$. Moreover $a = 0$ or $b = 0$; a and b are the exact base-adic depths; d and e are the valuation gcds of the corresponding base-free parts; and

$$\gcd(\gcd(d, e), |b - a|) = 1.$$

Since every counterexample has $\beta \geq 12$, these normalizations also satisfy the size dichotomy

$$w^d \geq 4096 \quad \text{or} \quad v^e \geq 3^{12}.$$

Status: [\[kernel-verified Lean\]](#) — `exists_simultaneous_primitive_output_normalization`, `IsLeastTwoBaseNonintegerSolution.simultaneousCoordinateGCD_eq_one`, and `exists_large_primitiveCore_of_not_alaogluErdosConjecture` in `OutputNormalization`. The gcd-one statement simultaneously controls the two outer power degrees and the difference of the two base-adic depths; it is stronger than merely excluding a common perfect-power degree.

The two primitive cores cannot simultaneously be the exceptional smooth pair $(w, v) = (3, 2)$. Indeed the two displayed output equations would make $\vartheta = \log 3 / \log 2$ a root of the nonzero quadratic

$$dT^2 + (a - b)T - e,$$

contradicting the kernel-verified transcendence of ϑ . Combined with Theorem 4.14, this gives an exact core-level dichotomy.

Corollary 5.8 (Canonical-core rationality dichotomy). *In the simultaneous normalization above,*

$$w^\beta \in \mathbb{Q} \iff w = 3, \quad v^\beta \in \mathbb{Q} \iff v = 2,$$

and these two rationality events cannot both occur. Hence at least one of w, v has a prime divisor outside $\{2, 3\}$, and at least one of w^β, v^β is nonrational. More intrinsically, for every nonintegral solution x , the two iterated outputs $(2^x)^x = 2^{x^2}$ and $(3^x)^x = 3^{x^2}$ cannot both be rational.

Status: [\[kernel-verified Lean\]](#) — `not_simultaneous_three_two_output_cores`, `exists_exact_rationality_classification_of_simultaneous_cores`, `exists_external_prime_in_simultaneous_primitive_output_cores`, and `exists_irrational_canonical_core_power_of_not_alaogluErdosConjecture` in `CoreIndependence`; the iterated-output statement is `TwoBaseNonintegerSolution.not_both_iterated_outputs_rational`. The packaged theorems retain the least solution, both primitive normalizations, and their exact output equations.

5.4 Exact counting

Theorem 5.9 (Exact unit-block count). *Assume a counterexample exists and let β be the primitive generator. For every $N \in \mathbb{N}_0$, the nonintegral solutions in $[N, N + 1)$ are precisely*

$$x_{N,k} = \lceil N - k\beta \rceil + k\beta, \quad 1 \leq k \leq \left\lfloor \frac{N+1}{\beta} \right\rfloor,$$

so their number is exactly $\lfloor (N+1)/\beta \rfloor$, and

$$\#(C \cap [2^N, 2^{N+1})) = 1 + \left\lfloor \frac{N+1}{\beta} \right\rfloor.$$

Proof. For fixed $k \geq 1$ there is at most one integer n with $N \leq n + k\beta < N + 1$; such $n \geq 0$ exists iff $k\beta < N + 1$, and then $n = \lceil N - k\beta \rceil$. Irrationality of β makes $(N+1)/\beta$ non-integral, giving the stated range; the one integral solution in the block is N . $\square$

Status: [\[kernel-verified Lean\]](#) — `ExactCounting` proves the explicit ceiling-image parametrization, `twoBaseNonintegerSolutionsInUnitBlock_ncard`, `twoBaseNaturalCandidatesInDyadicBlock_card`, and the exact finite-sum identity `cumulativeTwoBaseCandidateCount_eq`. `QuadraticAsymptotic` identifies that cumulative count exactly with the candidate count below 2^T and proves the sharp bounds and limit in (5).

With the verified bound $\beta > 12$ the block count is at most $\lfloor N/12 \rfloor$ (and at most $\lfloor N/20 \rfloor$ under the interval scan). Summing over blocks, the cumulative candidate count below 2^T is

$$C(T) = T + \sum_{r=1}^T \left\lfloor \frac{r}{\beta} \right\rfloor = \frac{T^2}{2\beta} + O(T). \quad (5)$$

More precisely, Lean proves

$$\frac{T(T+1)}{2\beta} \leq C(T) \leq \frac{T(T+1)}{2\beta} + T, \quad \frac{C(T)}{T^2} \rightarrow \frac{1}{2\beta}.$$

This is conditionally sharper than the verified universal bounds $((\log_2 N)^2$ and $\log_2 N \cdot \log_3 N$) and matches the verified conditional lower boxes in leading order: with $\beta > 12$ the leading coefficient is below $1/24$.

There is also an unconditional growth reformulation. The conjecture holds if and only if

$$\frac{\#(\mathcal{C} \cap [0, 2^T))}{T^2} \rightarrow 0.$$

Indeed, under the conjecture the numerator is exactly T ; under failure the preceding limit is the strictly positive constant $1/(2\beta)$. This is

`alaogluErdosConjecture_iff_candidate_count_below_two_pow_subquadratic.` [\[kernel-verified Lean\]](#)

5.5 Block Fourier averages and equidistribution

The fractional part of $x_{N,k}$ is $\{k\beta\}$: a high unit interval contains an initial segment of one irrational rotation orbit.

Theorem 5.10 (Finite Fourier tests on exact blocks). *Assume a counterexample exists and let β be its primitive generator. For every finitely supported coefficient function $c : \mathbb{Z} \rightarrow \mathbb{C}$, put*

$$P_c(t) = \sum_{h \in \mathbb{Z}} c(h) e^{2\pi i h t}, \quad K_N = \left\lfloor \frac{N+1}{\beta} \right\rfloor.$$

On the exact ceiling-parametrized points $x_{N,k} = \lceil N - k\beta \rceil + k\beta$,

$$\frac{1}{K_N} \sum_{k=1}^{K_N} P_c(x_{N,k}) \rightarrow c(0).$$

Status: [\[kernel-verified Lean\]](#) — `BlockWeyl` proves vanishing of every nonzero normalized Fourier mode on the exact block points. `BlockEquidistribution` proves `tendsto_primitiveUnitBlockTrigonometricAverage` for every finitely supported coefficient function, with limit `c 0`. This is finite trigonometric-polynomial convergence, not yet a theorem for arbitrary continuous tests.

Theorem 5.11 (Blockwise Weyl equidistribution). *Assume a counterexample exists. For every interval $I \subseteq [0, 1]$ whose boundary has measure zero,*

$$\frac{\#\{x \in (\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1) : \{x\} \in I\}}{\#\{(\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1)\}} \rightarrow |I| \quad (N \rightarrow \infty).$$

Proof. By Theorem 5.9 the relevant fractional parts are $\{\beta\}, \{2\beta\}, \dots, \{K_N\beta\}$ with $K_N = \lfloor (N+1)/\beta \rfloor \rightarrow \infty$. For every nonzero integer h the Weyl sum is a finite geometric series,

$$\frac{1}{K} \sum_{k=1}^K e^{2\pi i h k \beta} = \frac{e^{2\pi i h \beta} (1 - e^{2\pi i h K \beta})}{K(1 - e^{2\pi i h \beta})} \rightarrow 0,$$

since β is irrational. Weyl’s criterion [5] concludes. □

Status: [\[paper proof in this report\]](#) — the finite-trigonometric-polynomial input is kernel-verified by Theorem 5.10, but the Stone–Weierstrass/Weyl-criterion extension to all continuous tests and then to interval indicators has not been formalized. No effective discrepancy rate follows without further Diophantine input on β : the geometric-sum proof controls each fixed Fourier mode but not uniformly over many modes.

6 Equivalent one-number and radical formulations

6.1 Localization at the prime three

Theorem 6.1 (Localization reformulation). *Conjecture 2.1 is false if and only if there exists an odd integer $u > 1$ with*

$$u^\vartheta \in \mathbb{Z}[1/3],$$

equivalently, an odd $u > 1$ and $a \in \mathbb{N}_0$ with $3^a u^\vartheta \in \mathbb{N}$.

Proof. If x is a counterexample, write $2^x = 2^a u$ with u odd; since 2^x is not a power of two, $u > 1$. Using $(2^a)^\vartheta = 3^a$,

$$u^\vartheta = \frac{(2^x)^\vartheta}{(2^a)^\vartheta} = \frac{3^x}{3^a} \in \mathbb{Z}[1/3].$$

Conversely, if $u^\vartheta = B/3^a$ with $B \in \mathbb{N}$, set $x = a + \log_2 u$; then $2^x = 2^a u \in \mathbb{N}$ and $3^x = 3^a u^\vartheta = B \in \mathbb{N}$, and $x \notin \mathbb{Z}$ since otherwise the odd $u > 1$ would be a power of two. $\square$

Status: [\[kernel-verified Lean\]](#) —

`not_alaogluErdosConjecture_iff_exists_odd_localized_radical` and `alaogluErdosConjecture_iff_oddLocalizedRadicalExclusion` in `RadicalReformulation`; an equivalent candidate-level form, `alaogluErdosConjecture_iff_no_odd_multiple_candidate` together with the analytic $\mathbb{Z}[1/3]$ -form, is proved independently in `Localization`. The equivalence is kernel-verified; the localized-radical exclusion on its right-hand side remains the open conjecture. The allowance of 3-power denominators is exactly the freedom created by shifting the exponent by an integer.

6.2 Canonical radical algebraicity

In the notation of Theorem 5.3, $E = 3^{\log_2 w}$ satisfies $E^d = A/3^a \in \mathbb{Q}$. Minimality of d has an algebraic consequence.

Proposition 6.2 (Exact radical degree). *The binomial $X^d - A/3^a$ is irreducible over $\mathbb{Q}$, and E is algebraic of exact degree d .*

Proof. For a positive rational q , the binomial $X^d - q$ can be reducible only if q is a p -th power in $\mathbb{Q}$ for some prime $p \mid d$ (the additional $4 \mid d$ exceptional case concerns -4 times a fourth power and is irrelevant for $q > 0$). If $q = r^p$ with $p \mid d$, the positive real $E^{d/p}$ is the positive p -th root of q , hence rational, contradicting minimality of d . $\square$

Status: [\[kernel-verified Lean\]](#) —

`irreducible_X_pow_sub_C_of_least_rational_power` in `RadicalDegree` proves the general norm argument, and `oddCoreRpow_exact_radical_degree` specializes it to the least normalized radical index. It identifies the minimal polynomial with $X^d - A/3^a$ and proves its degree is exactly d .

Minimality of the same rational-power index imposes an additional arithmetic obstruction which is independent of irreducibility. Let $g(A) = \gcd_{\ell|A} v_\ell(A)$, with the empty-support convention $g(1) = 0$. Then

$$\gcd(d, \gcd(a, g(A))) = 1. \quad (6)$$

Indeed, a common divisor $k > 1$ would write $d = kd_0$, $a = ka_0$, and $A = R^k$; positivity would then give $E^{d_0} = R/3^{a_0} \in \mathbb{Q}$, contradicting the leastness of d . Primewise, if a prime p divides both d and a , at least one positive prime valuation of A is not divisible by p . This includes the edge cases $a = 0$ and $A = 1$ and does not require a separate reducedness assumption.

Status: [\[kernel-verified Lean\]](#) —

`rationalPowerIndex_gcd_denominator_valuationGCD_eq_one`, `exists_primeValuation_not_dvd_of_prime_dvd_rationalPowerIndex`, and the packaged `exists_exact_radical_degree_and_valuation_restriction` in `CanonicalRadicalValuation`.

This condition is sharp in the precise sense suggested by the proof. In reduced form, for every positive c ,

$$\exists b \in \mathbb{Q} : b^p = \frac{c}{3^a} \iff p \mid a \text{ and } p \mid g(c).$$

Consequently, for every $d > 0$, irreducibility of $X^d - c/3^a$ is equivalent to (6). The odd part is the rational binomial criterion. For the power of two in d , one iterates quadratic extensions: at the first lift a square would make $c/3^a$ a rational square, while at every later lift its norm is $-c/3^a < 0$, impossible for a square norm. Positivity is exactly what eliminates the classical exceptional negative fourth-power case. The pairwise strengthenings are false: $X^3 - 2/27$ is irreducible although $\gcd(3, 3) > 1$, while $X^3 - 8/3$ is irreducible although the numerator valuation $\gcd$ is divisible by 3. The canonical failure package also strengthens $c > 0$ to $c > 1$ and produces a prime divisor of c different from 3.

Status: [\[kernel-verified Lean\]](#) —

`exists_rat_pow_eq_reduced_iff_dvd_denominator_and_valuationGCD`, `irreducible_normalized_binomial_iff_valuationGCD_eq_one`, and `exists_exact_radical_with_nonthree_numerator_prime` in `CanonicalRadicalSharpness`.

The same normalized radical carries a further exact output restriction. Write $E^d = A/3^a$ in the reduced form above and retain the least solution β , so that $3^\beta = A$. Then

$$E \text{ is an algebraic integer} \iff a = 0, \quad (E^d)^\beta \in \mathbb{Q} \iff \exists u, v \in \mathbb{N}_0, A = 2^u 3^v.$$

When $a > 0$, reducedness gives $3 \nmid A$, and the exact statement sharpens to

$$(E^d)^\beta \in \mathbb{Q} \iff \exists u \in \mathbb{N}_0, \quad A = 2^u \quad \text{and} \quad (E^d)^\beta = w^{du} \in \mathbb{N}.$$

The algebraic-output determinant removes the apparent loss of the outer d -th power:

$$\text{Alg}_{\mathbb{Q}}(E^\beta) \iff \exists u, v \in \mathbb{N}_0, \quad A = 2^u 3^v.$$

When $a > 0$, this becomes the completely explicit statement

$$\text{Alg}_{\mathbb{Q}}(E^\beta) \iff \exists u \in \mathbb{N}_0, \quad A = 2^u \quad \text{and} \quad E^\beta = w^u \in \mathbb{N}.$$

Thus both $(E^d)^\beta$ and E^β have exact classifications; the latter is stronger and uses positivity to descend equality of d -th powers.

Status: [\[kernel-verified Lean\]](#) — `oddCoreRpow_isIntegral_iff_denominator_trivial`, `normalizedRadical_rpow_rational_iff_numerator_threeSmooth`, `normalizedRadical_rpow_rational_iff_exists_twoPow_and_eq_corePow`, and the packaged `exists_exact_radical_degree_and_thirdOutput_constraint` in `CanonicalRadicalThirdOutput`; and `oddCoreRpow_rpow_isAlgebraic_iff_numerator_threeSmooth`, `oddCoreRpow_rpow_isAlgebraic_iff_exists_twoPow_and_eq_corePow`, and the packaged `exists_canonical_radical_rpow_algebraic_constraint` in `CanonicalRadicalAlgebraicOutput`.

The exceptional smooth-numerator case for one radical cannot simply be discarded: the non-3 prime in A may be 2. The simultaneous normalization nevertheless gives a strict disjunction. Let

$$E = w^{\log_2 3}, \quad F = v^{\log_3 2},$$

for the primitive odd core w of 2^β and primitive 3-free core v of 3^β . Their rational powers are

$$E^d = \frac{3^b v^e}{3^a}, \quad F^e = \frac{2^a w^d}{2^b}.$$

At least one core has a prime divisor outside $\{2, 3\}$. On the opposite side this prime survives in the numerator of the displayed rational power. Cross-multiplying any hypothetical positive 2, 3-unit power then gives an impossible divisibility. Therefore

$$E \notin \sqrt[\infty]{\langle 2, 3 \rangle_{\mathbb{Z}}} \quad \text{or} \quad F \notin \sqrt[\infty]{\langle 2, 3 \rangle_{\mathbb{Z}}},$$

and consequently E^β or F^β is transcendental. This is an all-positive-powers exclusion on one side, stronger than merely knowing that one selected output is nonrational.

Status: [\[kernel-verified Lean\]](#) — `not_hasPositiveTwoThreeUnitPower_of_rational_power_external_prime`, `simultaneous_core_radical_power_normalization`, and the packaged `exists_simultaneous_canonical_radical_divisibleHull_exclusion` in `CanonicalRadicalDivisibleHull`.

A counterexample therefore produces the very special algebraic number $E = 3^{\log_2 w}$, reached from a transcendental exponent. Taking logarithms of $E^d = A/3^a$:

$$d \log w \log 3 = (\log A - a \log 3) \log 2. \quad (7)$$

This is a *bilinear* relation among logarithms of integers. Baker’s theorem controls linear forms in logarithms with algebraic coefficients [3]; it does not rule out (7). The four exponentials conjecture does. The equation is a precise description of the transcendence-theoretic wall, not a disguised elementary factorization problem.

The exact radical field makes the same wall visible algebraically. If $q = A/3^a$ and E has minimal polynomial $X^d - q$, then in $\mathbb{Q}(E)$

$$N_{\mathbb{Q}(E)/\mathbb{Q}}(E) = (-1)^d(-q), \quad \text{Tr}_{\mathbb{Q}(E)/\mathbb{Q}}(E) = 0 \quad (d > 1).$$

These are the full product and sum of the conjugates, not estimates. They do not create a new exponential corner: embeddings replace E by roots ζE but do not act on the transcendental exponent β .

Nor does the symmetric radical F supply an independent third base. The normalization itself gives

$$2^b F^e = 2^a w^d,$$

an explicit collision among nonnegative monomials in $2, w, F$. The corresponding output identity $3^a E^d = 3^b v^e$ is simply its $\log_2 3$ -multiple. Thus the most tempting use of the new algebraic six-exponentials theorem fails its essential monomial-injectivity hypothesis for a structural, kernel-verified reason.

Status: [\[kernel-verified Lean\]](#) —

swappedRadical_relation_blocks_monomial_injectivity,
simultaneousNormalization_blocks_swapped_monomial_injectivity,
exactRadical_adjoin_norm_gen, exactRadical_adjoin_trace_gen_eq_zero, and
oddCoreRpow_exact_field_norm_and_trace in CanonicalRadicalFieldNorm.

6.3 Where the wall stands in the smallest case

The smallest structural question already meets the open core. A 3-smooth candidate $m = 2^a 3^b$ ($b \geq 1$) would require $3^a E_3^b \in \mathbb{N}$ for

$$E_3 = 3^{\log_2 3} = e^{(\log 3)^2 / \log 2} = 5.7045224946911176 \dots,$$

and even the irrationality of E_3 is open: it is a “three-logarithms” quantity outside the reach of Gelfond–Schneider, Baker’s method, and the six exponentials theorem. No currently known technique excludes odd core $w = 3$; this is why the corpus emphasizes counting, structure, and verified finite regions rather than infinite families of exclusions, and why Conjecture 6.3 below is the honest name of the obstruction.

Conjecture 6.3 (Radical four-exponentials target). *For every odd integer $u > 1$, $u^{\log_2 3} \notin \mathbb{Z}[1/3]$. Equivalently, for every primitive odd $w > 1$ and every $d \geq 1$, $(3^{\log_2 w})^d \notin \mathbb{Z}[1/3]$.*

Status: [\[kernel-verified Lean\]](#) — `PrimitiveRadical` proves unique primitive-power decomposition of every integer $u > 1$ and the exact equivalence `alaogluErdosConjecture_iff_primitiveOddLocalizedRadicalExclusion`. Thus proper-power choices of u introduce no genuinely new cases.

By Theorem 6.1 both formulations are exactly equivalent to Conjecture 2.1. A proof of the special case $u = 3$ — even of the weaker assertion that $3^{\log_2 3}$ is irrational — by any genuinely new mechanism would already be a meaningful breakthrough.

7 Sharper finite-difference obstructions

The earlier local theorem (Theorem 4.5) forbids three candidates in progression under the convenient condition $h^5 \leq n$. The sharp arbitrary-order curvature theorem is now kernel-verified for every $r \geq 2$, including the three- and four-term special cases.

For a function f and step $h > 0$ write $\Delta_h f(t) = f(t+h) - f(t)$ and $\Delta_h^r = \Delta_h \circ \cdots \circ \Delta_h$. Iterating the fundamental theorem of calculus,

$$\Delta_h^r f(n) = \int_{[0,h]^r} f^{(r)}(n + t_1 + \cdots + t_r) dt_1 \cdots dt_r. \quad (8)$$

For $f(t) = t^\vartheta$ and $r \geq 2$, $f^{(r)}(t) = \vartheta(\vartheta-1)\cdots(\vartheta-r+1)t^{\vartheta-r}$; since $1 < \vartheta < 2$ this has a fixed nonzero sign on $(0, \infty)$ and its absolute value decreases in t .

Theorem 7.1 (Higher-difference progression obstruction). *Let $r \geq 2$, $n, h \in \mathbb{N}_{>0}$, and $c_r = |\vartheta(\vartheta-1)\cdots(\vartheta-r+1)|$. If all of $n, n+h, \dots, n+rh$ are candidates, then $1 \leq c_r h^r n^{\vartheta-r}$. Such a progression is therefore impossible whenever*

$$c_r h^r < n^{r-\vartheta}. \quad (9)$$

Proof. If all $n+jh$ are candidates, every $(n+jh)^\vartheta$ is an integer, so the integer combination $\Delta_h^r f(n)$ is an integer; by (8) and the fixed sign of $f^{(r)}$ it is nonzero, hence of absolute value at least one. Bounding $|f^{(r)}|$ by its value at n ,

$$1 \leq |\Delta_h^r f(n)| \leq c_r h^r n^{\vartheta-r}. \quad \square$$

Status: [\[kernel-verified Lean\]](#) — `HigherDifference` proves the generic iterated mean-value identity `exists_iterated_fwdDiff_eq_pow_mul`, its real-power specialization `exists_iterated_fwdDiff_rpow_point`, integrality of the binomial forward difference, and `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` uniformly for every $r \geq 2$. The displayed iterated-integral identity (8) is not used by the Lean proof; the formal argument instead iterates the mean-value theorem.

For $r = 2$ this gives the sharp three-term criterion

$$\vartheta(\vartheta-1)h^2 < n^{2-\vartheta}, \quad \text{i.e.} \quad h < 1.038547802 \dots \cdot n^{0.2075187 \dots}, \quad (10)$$

strictly weaker as a hypothesis than $h \leq n^{1/5}$; this sharp criterion is the theorem `second_difference_logThreeDivLogTwo_ap_of_curvature` in `SharperProgressions`. For $r = 3$, `ThirdDifference` kernel-checks the exact third-difference identity and the corresponding four-candidate obstruction. The first thresholds $h < C_r n^{1-\vartheta/r}$, $C_r = c_r^{-1/r}$:

r	length $r + 1$	exponent $1 - \vartheta/r$	C_r
2	3	0.207518750	1.038547802
3	4	0.471679166	1.374849673
4	5	0.603759375	1.164124485
5	6	0.683007500	0.946705202
6	7	0.735839583	0.778537835
7	8	0.773576786	0.652646038
8	9	0.801879687	0.557368999

As r grows the admissible step exponent approaches one. This looks tantalizing but does not contradict the exact candidate monoid: in a dyadic block near X the conditional number of candidates is $O(\log X)$, so the average additive gap is of order $X/\log X$, which for each fixed r exceeds $X^{1-\vartheta/r}$. Long-progression leverage would need r growing with the height together with control of the rapidly growing c_r and a combinatorial source of such progressions.

8 Independent finite verification through 2^{20}

Proposition 8.1 (Finite exclusion under interval-software trust). *For every integer $1 \leq m < 2^{20}$: if $m^\vartheta \in \mathbb{Z}$ then m is a power of two. Equivalently, $2^x, 3^x \in \mathbb{Z}$ and $2^x < 2^{20}$ imply $x \in \mathbb{Z}$; a nonintegral solution satisfies $x > 20$.*

Status: [\[interval computation\]](#) — CPython 3.13.5, mpmath 1.3.0 interval arithmetic at 40 decimal digits, outward endpoints. Reproducible but *not* replayed by the Lean kernel; not interchangeable with `twelve_le_of_not_integer_of_two_three_rpow_integer`.

For each non-power of two m , a floating approximation proposes $n = \lfloor m^\vartheta \rfloor$; the proposal is then *certified* by interval proofs of the two strict inequalities

$$\log m \log 3 - \log n \log 2 > 0, \quad (11)$$

$$\log(n + 1) \log 2 - \log m \log 3 > 0, \quad (12)$$

which together say exactly $n < m^\vartheta < n + 1$. Once both interval lower endpoints are positive the floating proposal has no remaining logical role. All 1,048,555 non-powers of two below 2^{20} passed, in two ranges:

Range	Checked	Failures	Smallest certified log margins
$1 \leq m < 2^{19}$	524,268	0	lower 1.4147×10^{-15} at $(m, n) = (189427, 231498127)$; upper 3.0841×10^{-15} at $(306982, 497586055)$.
$2^{19} \leq m < 2^{20}$	524,287	0	lower 9.4902×10^{-16} at $(958460, 3023921230)$; upper 2.3020×10^{-15} at $(556061, 1275861767)$.

At the globally tightest case the 40-digit interval had width about 1.3×10^{-54} , some thirty-nine orders of magnitude below the certified positive margin; a separate 100-digit computation gives $958460^9 - 3023921230 = 4.14020498680340526 \times 10^{-6}$. The verification script, full outputs, and SHA-256 manifests are reproduced in Appendix D.

From scan to kernel certificate. The repository’s own architecture (section 4.2) shows the stronger trust model: generate rational certificates externally, revalidate them independently, and replay only exact integer comparisons in Lean by `decide`. A scalable 2^{20} formal certificate should share convergent enclosures across ranges, chunk the data modules to bound elaboration memory, and rely on verified binary powering rather than giant literals; the worst cases identified by the scan (the four extremal pairs above) predict the required tier depth.

9 Attempts at a full proof and their exact failure points

9.1 Collapse rank two to rank one

The rank argument leaves a perfectly coherent rank-two possibility, now known exactly: $\mathcal{C} = \langle 2, M \rangle$. Nothing in the rank machinery distinguishes this from the allowed monoid of powers of two; excluding the second generator *is* the original transcendence problem.

9.2 Density of multiples against endpoint gaps

At paper level the orbit $k\beta \bmod 1$ is equidistributed (Theorem 5.11); the finite-Fourier convergence needed for Weyl’s criterion is kernel-verified by Theorem 5.10. The integrality gap at height k is of exponential scale M^{-k} (Theorem 4.2). Equidistribution controls fixed intervals — and with effective discrepancy, polynomially shrinking ones along subsequences — but numbers with bounded partial quotients show that equidistributed multiples can avoid exponentially small targets forever. Closing this route needs Diophantine information specific to the transcendental β with $2^\beta, 3^\beta \in \mathbb{N}$, not generic metric theory.

9.3 Exact counts against local progression obstructions

Conditional failure gives $O(\log X)$ candidates per multiplicative block — far below the positive-density regime of Roth-type theorems — while Theorem 7.1 reaches steps $X^{1-\vartheta/r}$ against average gaps $X/\log X$. Sparse exact structure and local curvature are mutually consistent. A three-term progression of candidates would solve the S -unit equation $2^{n_1}M^{k_1} + 2^{n_3}M^{k_3} = 2^{n_2+1}M^{k_2}$; general S -unit theory yields finiteness statements, not the required nonexistence.

9.4 Exploit the primitive output pair

The pair (M, A) admits no matched depth, no common perfect power, and no nontrivial affine decomposition (Corollaries 5.4, 5.5 and 5.6). What remains is exactly relation (7), a product of logarithms outside the scope of linear-forms machinery.

9.5 Force algebraicity of a forbidden auxiliary exponential

The five exponentials theorem [10, 6] makes $e^{\gamma x}$ transcendental for every nonzero algebraic γ , given the four algebraic corner exponentials. The natural auxiliary quantities $p^x = e^{x \log p}$ have transcendental coefficient $\log p$, so the theorem does not directly convert divisors of M or A into a contradiction. The six exponentials theorem shows b^x is transcendental for any positive algebraic base b multiplicatively independent of 2, 3; the committed Theorem 4.14 is its formalized rational-level shadow for positive rational bases. It proves $b^x \notin \mathbb{Q}$ rather than the classical theorem’s stronger transcendence conclusion. Neither prevents b from dividing the integer 2^x ; these statements control the distinct quantity b^x .

9.6 The linear-form-in-log-products formulation of the wall

The obstruction can be given one further exact name. A counterexample is a pair of integers $m, A \geq 2$ with $\log m \log 3 = \log A \log 2$. Expanding both integers over their prime factorizations $m = \prod_p p^{m_p}$, $A = \prod_p p^{a_p}$, failure of the conjecture is equivalent to the nontrivial vanishing of

$$D = \sum_p (m_p \log 3 - a_p \log 2) \log p,$$

an *integer-coefficient linear form in the products of two prime logarithms* $\{\log p \cdot \log 2, \log p \cdot \log 3\}$. Baker’s theory bounds linear forms in logarithms with *algebraic* coefficients; here every coefficient of $\log p$ is itself a transcendental logarithm, and no lower-bound technology exists for such forms. Indeed even the $\mathbb{Q}$ -linear independence of the family $\{\log p \cdot \log q : p < q \text{ prime}\}$ is an open problem (it follows from, and is essentially a fragment of, the four exponentials circle of conjectures). Every route examined in this report — determinants, product formulas in the radical field $\mathbb{Q}(E)$, auxiliary exponentials, equidistribution against shrinking targets — terminates at a special case of exactly this independence statement. A genuine proof therefore requires either a fundamentally new lower bound for linear forms in log-products, or a mechanism exploiting the positivity and integrality of the coefficients (m_p, a_p) that has no analogue in the general four-exponentials setting.

10 Consolidated research program

The two source documents proposed overlapping programs; the exact structure theory settles some directions and re-prioritizes the rest. Ordered by leverage and feasibility:

Priority A: finish the exact conditional consequences

The primitive normalization, uniqueness of the odd core and its coordinates, least non-integer solution and affine primitivity, simultaneous two-output normalization, rational-power index, denominator support, and exact primitive-generator theorem are complete in `PrimitiveOddCore`, `LeastSolution`, `OutputNormalization`, `ThreeDenominatorNormalization`, `RationalPowerIndex`, and `PrimitiveGenerator`. `ExactCounting` and `QuadraticAsymptotic` prove the exact unit-block and dyadic counts, the cumulative finite sum, its sharp quadratic limit, and the subquadratic-growth equivalence. `BlockWeyl` and `BlockEquidistribution` prove every fixed Fourier mode and finite trigonometric polynomial. `CoreIndependence` completes

the rationality classification of the two simultaneous primitive cores: at least one core has an external prime, and at least one of their β -th powers is nonrational. The remaining work in this priority is the functional-analytic extension to arbitrary continuous tests and interval indicators in Theorem 5.11. This supersedes the earlier “exponent semigroup” direction: the kernel-verified classification shows there are no Frobenius-type gaps to study.

Priority B: exploit the uniform higher-difference theorem

`HigherDifference` now formalizes a generic iterated forward-difference mean-value identity and proves Theorem 7.1 uniformly for every $r \geq 2$. The remaining problem is no longer analytic formalization but combinatorial leverage: propagate the sharp variable-order criterion through sparse-set arguments, with r allowed to grow with height, and determine whether the rapidly growing falling coefficient still permits a contradiction with the exact conditional monoid. For concrete finite verification, the rational effective three-term criterion $h^{400} \leq n^{83}$ (`not_three_arithmetic_progression_twoBaseNaturalCandidates_sharp` in `SharpProgression`, with the kernel-replayed enclosure $\vartheta < 317/200$ from $3^{200} < 2^{317}$) complements the exact-curvature form.

Priority C: kernel-replay the 2^{20} region

The scan shows the finite truth is computationally routine; the bottleneck is a compact kernel-replayable proof object. Adaptive convergent enclosures, chunked certificate modules, and shared tiers along ranges are the ingredients; the target theorem is the 1048576 analogue of Theorem 4.3. Every descent constant then improves to 20 by bookkeeping.

Priority D: use the exact radical field

`RadicalReformulation` now formalizes Theorem 6.1 and its symmetric base-swapped version, `PrimitiveRadical` reduces it canonically to power-primitive odd bases, while `PrimitiveGenerator` and `ThreeDenominatorNormalization` supply the primitive odd w , least index d , and reduced $q \in \mathbb{Z}[1/3]$ with $(3^{\log_2 w})^d = q$. `RadicalDegree` proves Proposition 6.2, including the exact minimal polynomial. `CanonicalRadicalValuation` proves the common-power obstruction (6) and its primewise form; `CanonicalRadicalSharpness` identifies the exact rational-power criterion, proves the all-degree irreducibility equivalence, and forces a non-3 prime in the canonical numerator. `CanonicalRadicalThirdOutput` also characterizes algebraic integrality of the radical and characterizes exactly when $(E^d)^\beta$ is rational in terms of the reduced numerator; `CanonicalRadicalAlgebraicOutput` upgrades this to an exact classification of algebraicity of E^β itself; `CanonicalRadicalDivisibleHull` proves that in the simultaneous normalization at least one of the two radicals lies outside the entire positive 2, 3-unit divisible hull; `CanonicalRadicalFieldNorm` computes the exact norm and trace and exhibits the monomial collision that blocks a direct three-base use of the algebraic six-exponentials theorem, while `CoreIndependence` rules out simultaneous smooth primitive cores. The next target is a genuinely new number-field argument using this explicit extension; the radical exclusion itself remains equivalent to, and exactly as open as, the original conjecture.

Priority E: five and sharp six exponentials

Formalizing the five exponentials theorem or Roy’s sharp six exponentials theorem [6] would add restrictions such as transcendence of $e^{\gamma x}$ for algebraic $\gamma \neq 0$. The effort is comparable to the completed Gelfond–Schneider port and should be undertaken with a specific downstream lemma in mind. The rational arithmetic analogue is now complete: **RationalSixExponentials** treats any three multiplicatively independent positive rational bases and three rational outputs; **RationalThirdBase** handles a denominator-unrestricted rational output at a natural third base under monomial injectivity; and **RationalBaseThird** handles both a rational base and rational output, with exact 2,3-unit classification. The genuinely stronger algebraic-third-output step is now complete for natural third bases: **AlgebraicOutputBridge** clears an arbitrary algebraic output into a number-field integer, **AlgebraicIntegerDeterminant** supplies the norm lower bound across all conjugates, and **AlgebraicThirdBase** absorbs the resulting height and degree loss in the same interpolation margin. The positive-rational-base extension is complete as well: **AlgebraicRationalBase** proves that, for a hypothetical nonintegral solution, q^x is algebraic exactly for rational 2,3-units. Finally, **FiniteAlgebraicOutputBridge** and **AlgebraicSixExponentials** treat three arbitrary positive algebraic real bases with injective nonnegative monomials and three algebraic outputs, allowing bases below one and negative exponents. **AlgebraicBaseUnitPower** extracts the exact divisible-hull description of the dependent algebraic-output locus. What remains beyond ordinary six exponentials is to formalize a genuinely sharp five/six-exponentials interface with a new downstream use.

Priority F: auxiliary-function methods tailored to integrality

A route beyond four exponentials must exploit more than algebraicity: here all corner values are positive integers, all their powers remain integers, and the primitive pair is canonical. Promising variants: use several non-Archimedean valuations simultaneously; use Corollary 5.6 to prevent determinant factors from sharing perfect powers; work with norms in the explicit radical field $\mathbb{Q}(E)$ of Proposition 6.2, combining Archimedean smallness with denominator support only at 3; search for a product-formula contradiction across all places of $\mathbb{Q}(E)$. The last is the most structurally aligned with the radical formulation.

Priority G: primitive-pair search instead of raw candidate scanning

A counterexample has a primitive pair (M, A) with $M^\vartheta = A$, $\min(v_2 M, v_3 A) = 0$, no common perfect power, and no matched affine decomposition. Finite searches can reject candidates early on these conditions and record near misses keyed by odd core, revealing whether closest approaches cluster by smoothness pattern, denominator, or rational-power index d .

Priority H: quantitative irrationality and discrepancy

Effective irrationality measures for ϑ (Páde/hypergeometric methods) would replace per-value certificate tiers by a uniform polynomial criterion and likely extend verified regions by orders of magnitude. They do not directly control the generator $\beta = \log_2 M$ for unknown M ; that would require an effective measure derived from the relation (7), which lies beyond ordinary linear forms in logarithms.

Downstream: the colossally abundant interface

Formalizing the original application — the conjecture’s rational-power form implies that quotients of consecutive colossally abundant numbers are prime [1] — connects the corpus to its historical source. It is logically downstream and should follow the core structure modules.

11 Conclusions

The repository contains a broad, kernel-verified attack on the Alaoglu–Erdős conjecture whose themes are complementary: interpolation determinants settle three independent bases; Gelfond–Schneider excludes algebraic irrational exponents; exact certificates create a growing zero-free region; unique factorization yields a canonical primitive odd core and an exact conditional solution monoid; arbitrary-order forward differences force sharp local additive sparsity; leastness gives exact affine primitivity; and unconditional equivalences with fractional-part gaps and localized radicals identify the entire remaining distance to the conjecture.

The conjecture does not fall to a combination of these results, and the reason can now be stated exactly. A counterexample would not generate a messy exceptional set that density or semigroup arguments might squeeze; it would generate the perfectly rigid free monoid

$$\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta, \quad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\},$$

whose block counts, fractional parts, and multiplicative relations are explicit and internally consistent with every verified gap and density bound. The remaining assertion is that the second generator cannot exist — equivalently (Conjecture 6.3), that no odd $u > 1$ has $u^{\log_2 3} \in \mathbb{Z}[1/3]$.

The free-monoid classification, simultaneous primitive output normalization, exact block counts, sharp cumulative asymptotic, finite-trigonometric-polynomial block convergence, primitive localized-radical reduction, exact radical degree, denominator-unrestricted rational-output determinants under monomial injectivity, the exact natural- and rational-base algebraic-output classifications, the divisible-hull classification for all positive algebraic bases, the rank-one second-iterate output locus, the canonical-radical E^β classification and simultaneous divisible-hull exclusion, the exact radical-field norm/trace and swapped-base rank obstruction, and the uniform higher-difference obstruction are kernel-verified. Full blockwise equidistribution for arbitrary continuous tests and intervals remains a paper theorem. What remains at the transcendence frontier is not denominator clearing or a number-field determinant lower bound, but the two-base/four-exponentials boundary itself.

That reformulation is progress rather than pessimism: it removes artificial degrees of freedom, supplies a canonical primitive object, yields exact conditional counting statements, and gives a focused interface for future transcendence methods. The near-term program is listed in [section 10](#); a complete proof still requires a new idea at the four-exponentials boundary, but the target is now narrow and the surrounding formal infrastructure is strong enough to test such an idea precisely.

A Status matrix for major statements

Statement	Status	Location
$2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$	[kernel-verified Lean]	IntegerExponent
Nonintegral solution irrational, transcendental	[kernel-verified Lean]	TwoBaseIntegerExponent, Transcendence
$2^x < 256 \Rightarrow x \in \mathbb{Z}$	[kernel-verified Lean]	FiniteCheck256
$2^x < 4096 \Rightarrow x \in \mathbb{Z} \ (x \geq 12)$	[kernel-verified Lean]	FiniteCheck4096
Rank of candidate exponent vectors ≤ 2	[kernel-verified Lean]	MultiplicativeRank
Common odd core $2^i w^j$ (existence)	[kernel-verified Lean]	OddCore
Counting $\log_2 N \cdot \log_3 N$; linear dyadic blocks	[kernel-verified Lean]	OddCore
Density dichotomy; gap equivalences; infinitude	[kernel-verified Lean]	FractionalPartDensity
3-smooth classification of further bases	[kernel-verified Lean]	ThreeSmooth
Affine descent, $r + 8k \leq x$	[kernel-verified Lean]	ArithmeticRigidity
Three-term AP obstruction under $h^5 \leq n$	[kernel-verified Lean]	ZeroDensity
Shrinking-target Diophantine bounds	[kernel-verified Lean]	ShrinkingTarget
Gelfond–Schneider theorem	[kernel-verified Lean]	GelfondSchneider
4EC $\Rightarrow$ conjecture	[kernel-verified Lean]	FourExponentialsReduction
Primitive (canonical) odd core; unique coordinates	[kernel-verified Lean]	PrimitiveOddCore
Unique least solution; affine primitive output pair	[kernel-verified Lean]	LeastSolution
Simultaneous primitive output normalization; gcd one; size dichotomy	[kernel-verified Lean]	OutputNormalization
Power-of-three denominator normalization	[kernel-verified Lean]	ThreeDenominatorNormalization
Rational-power index and exact solution monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$	[kernel-verified Lean]	RationalPowerIndex, PrimitiveGenerator
Exact block counts and cumulative finite sum	[kernel-verified Lean]	ExactCounting
Cumulative $T^2/(2\beta) + O(T)$ asymptotic and subquadratic iff	[kernel-verified Lean]	QuadraticAsymptotic
Exact block averages for finite trigonometric polynomials	[kernel-verified Lean]	BlockWeyl, BlockEquidistribution
Full continuous-test/interval blockwise equidistribution	[paper proof in this report]	Theorem 5.11
Generic rational six-exponentials theorem for three bases	[kernel-verified Lean]	RationalSixExponentials
Generic algebraic six-exponentials theorem for positive real bases	[kernel-verified Lean]	AlgebraicSixExponentials
Denominator-unrestricted rational output under monomial injectivity	[kernel-verified Lean]	RationalThirdBase

Statement	Status	Location
Positive rational-base 2, 3-unit classification	[kernel-verified Lean]	RationalBaseThird
Positive rational-base algebraic-output classification	[kernel-verified Lean]	AlgebraicRationalBase
Positive algebraic-base divisible-hull classification	[kernel-verified Lean]	AlgebraicBaseUnitPower
Rank-three positive-algebraic-output-locus equivalence	[kernel-verified Lean]	AlgebraicRankThreeEquivalence
Number-field determinant bound and finite-family algebraic-output bridge	[kernel-verified Lean]	AlgebraicIntegerDeterminant, AlgebraicOutputBridge, FiniteAlgebraicOutputBridge
Natural-base algebraic-output classification and equivalent formulations	[kernel-verified Lean]	AlgebraicThirdBase, AlgebraicConsequences
Mixed-output half-plane and ratio-locus rank-two equivalence	[kernel-verified Lean]	AlgebraicOutputLocus
Canonical-core rationality dichotomy and external prime	[kernel-verified Lean]	CoreIndependence
Sharp three-term and four-term AP obstructions ($r = 2, 3$)	[kernel-verified Lean]	SharperProgressions, ThirdDifference
Uniform higher-order AP obstruction ($r \geq 2$)	[kernel-verified Lean]	HigherDifference
Localized-radical and symmetric reformulations	[kernel-verified Lean]	RadicalReformulation
Primitive localized-radical reduction	[kernel-verified Lean]	PrimitiveRadical
Exact radical degree and binomial irreducibility	[kernel-verified Lean]	RadicalDegree
Least-index denominator/numerator valuation gcd restriction	[kernel-verified Lean]	CanonicalRadicalValuation, CanonicalRadicalSharpness
Normalized-radical integrality and rational-output restrictions	[kernel-verified Lean]	CanonicalRadicalThirdOutput
Canonical radical E^β algebraic-output classification	[kernel-verified Lean]	CanonicalRadicalAlgebraicOutput
Simultaneous canonical-radical divisible-hull exclusion	[kernel-verified Lean]	CanonicalRadicalDivisibleHull
Canonical radical norm, trace, and swapped-base rank obstruction	[kernel-verified Lean]	CanonicalRadicalFieldNorm
Effective rational three-term criterion ($h^{400} \leq n^{83}$)	[kernel-verified Lean]	SharpProgression
Candidate-form localization equivalence	[kernel-verified Lean]	Localization
Convergent-denominator growth $q_{n+1} < 2m^{q_n}$	[kernel-verified Lean]	DenominatorGrowth
No nonintegral candidate with $2^x < 2^{20}$	[interval computation]	Proposition 8.1
Full Alaoglu–Erdős conjecture	open	reduced to Conjecture 6.3

B Formalization inventory

All modules live under `Analysis/ExponentialIdentities/Lean/ExponentialIdentities/`, in namespace `LeanProofs`. No sorry, no extra axioms, no `native_decide`; certificate tables are replayed by the kernel.

Module	Contents
IntegerExponent (+3 modules)	three-base theorem, interpolation determinants
TwoBaseIntegerExponent	conjecture statement, candidates, gaps, $2^x < 10$
.../FiniteCheck, FiniteCheck256	zero-free regions $2^x < 100$, $2^x < 256$
.../FiniteCheck4096	zero-free region $2^x < 4096$ (kernel decide table)
.../ZeroDensity	AP obstruction, Roth-number counting, $o(N)$
.../NonintegerDensity	nonintegral candidate refinement, density zero
.../MultiplicativeRank	rank ≤ 2 , ϑ irrational, $(\log_2 N)^2$
.../CandidateStructure	cross-valuation identity, conditional box growth
.../OddCore	common odd core, $\log_2 N \log_3 N$, dyadic blocks
.../PrimitiveOddCore	gcd-normalized canonical core, unique coordinates
.../LeastSolution	unique positive least solution, exact affine primitivity
.../ThreeDenominatorNormalization	reduced denominator supported at 3
.../RationalPowerIndex	least rational-power index, exact solution monoid
.../PrimitiveGenerator	canonical primitive generator and unique coordinates
.../ExactCounting	exact unit/dyadic block counts and cumulative finite sum
.../QuadraticAsymptotic	sharp quadratic limit; subquadratic-growth equivalence
.../BlockWeyl	exact-block geometric sums and vanishing nonzero Fourier modes
.../BlockEquidistribution	finite trigonometric-polynomial block averages
.../OutputNormalization	simultaneous primitive outputs, gcd one, size dichotomy
.../RationalSixExponentials	generic rational three-base determinant theorem
.../RationalThirdBase	denominator-unrestricted rational output under monomial injectivity
.../RationalBaseThird	exact positive-rational-base 2, 3-unit classification
.../AlgebraicRationalBase	exact positive-rational-base algebraic-output classification
.../AlgebraicBaseUnitPower	divisible-hull classification for positive algebraic bases
.../AlgebraicRankThreeEquivalence	rank-three algebraic-output-locus equivalence
.../AlgebraicIntegerDeterminant	number-field norm and conjugate determinant bounds
.../AlgebraicOutputBridge	finite-number-field realization and integral scaling
.../FiniteAlgebraicOutputBridge	shared number field and denominator for a finite algebraic family
.../AlgebraicSixExponentials	generic positive-real algebraic six-exponentials determinant
.../AlgebraicThirdBase	algebraic output at an independent natural third base
.../AlgebraicConsequences	exact natural-base classification, iterates, equivalent conjectures
.../AlgebraicOutputLocus	monomial half-plane, mixed iterates, rank-one ratio locus
.../CoreIndependence	canonical-core rationality dichotomy, external prime
.../RadicalReformulation	localized-radical equivalences, base-swapped form
.../PrimitiveRadical	canonical primitive-power reduction of radical target
.../RadicalDegree	irreducible binomial, exact minpoly and degree
.../CanonicalRadicalValuation	least-index gcd and primewise valuation restrictions
.../CanonicalRadicalSharpness	exact power criterion, all-degree irreducibility, numerator prime
.../CanonicalRadicalThirdOutput	radical integrality and rational-output restrictions
.../CanonicalRadicalAlgebraicOutput	exact algebraic-output classification for E^β
.../CanonicalRadicalDivisibleHull	simultaneous all-positive-powers hull exclusion
.../CanonicalRadicalFieldNorm	exact norm/trace and swapped-radical monomial collision
.../SharperProgressions	sharp three-candidate curvature obstruction
.../ThirdDifference	exact third difference, four-candidate obstruction
.../HigherDifference	uniform arbitrary-order forward-difference obstruction
.../ArithmeticRigidity	descent, $r + 8k \leq x$
.../ThreeSmooth	3-smooth classification of further bases
.../ShrinkingTarget	fract shrinking targets, Diophantine lower bounds
.../DenominatorGrowth	convergent-denominator growth cap $Q < 2m^q$
.../Transcendence	GS applications: transcendence of solutions, of $\log_2 3$
.../FractionalPartDensity	density dichotomy, gap equivalences, infinitude
.../Localization	candidate-form localization equivalence
.../SharpProgression	effective rational three-term criterion
.../FourExponentialsReduction	4EC $\Rightarrow$ conjecture
GelfondSchneider/...	Gelfond–Schneider theorem (port of Karataakis)

C A symmetric formulation

Everything above has a base-swapped version. Put $\eta = \log_3 2 = 1/\vartheta$. Failure of the conjecture is also equivalent to the existence of an integer $v > 1$ with $3 \nmid v$ and

$$v^\eta \in \mathbb{Z}[1/2].$$

Status: [\[kernel-verified Lean\]](#) —

`not_alaogluErdosConjecture_iff_exists_threeFree_localized_radical` and
`alaogluErdosConjecture_iff_threeFreeLocalizedRadicalExclusion` in
`RadicalReformulation`.

For the primitive pair (M, A) , factoring A into a power of 3 times a 3-coprime part produces a symmetric primitive core and rational-power index; the two normalizations give the same least exponent β , and their compatibility condition is exactly $\log M \log 3 = \log A \log 2$. The symmetry is useful in formalization — some divisibility arguments are easier on one side — and confirms that Corollary 5.4 is not an artifact of privileging the base 2.

D Interval computation: script and outputs

```

1 import math, time, hashlib
2 import mpmath as mp
3
4 START = 1
5 LIMIT = 1 << 19
6 DPS = 40
7
8 mp.iv.dps = DPS
9 ln2 = mp.iv.log(2)
10 ln3 = mp.iv.log(3)
11 theta_float = math.log(3.0) / math.log(2.0)
12
13 checked = 0
14 failures = []
15 min_lower = (float("inf"), None, None)
16 min_upper = (float("inf"), None, None)
17 manifest = hashlib.sha256()
18 start_time = time.time()
19
20 for m in range(START, LIMIT):
21     if (m & (m - 1)) == 0:      # integral exponent: m is a power of two
22         continue
23
24     # Only proposes a neighboring integer. The interval tests below certify it.
25     n = math.floor(math.exp(theta_float * math.log(m)))
26
27     lhs = mp.iv.log(m) * ln3
28     lower_diff = lhs - mp.iv.log(n) * ln2
29     upper_diff = mp.iv.log(n + 1) * ln2 - lhs
30
31     lower_lb = float(lower_diff.a)
32     upper_lb = float(upper_diff.a)

```

```

33     if not (lower_lb > 0.0 and upper_lb > 0.0):
34         failures.append((m, n, str(lower_diff), str(upper_diff)))
35         break
36
37     min_lower = min(min_lower, (lower_lb, m, n))
38     min_upper = min(min_upper, (upper_lb, m, n))
39     manifest.update(f"{m}:{n}:".encode())
40     checked += 1
41
42 print("checked_nonpowers=", checked)
43 print("failures=", len(failures))
44 print("min_lower_log_margin=", min_lower)
45 print("min_upper_log_margin=", min_upper)
46 print("sha256_m_n_manifest=", manifest.hexdigest())
47 print("elapsed_seconds=", time.time() - start_time)
48 assert not failures

```

Listing 1: Interval verification for one range. Set START to 1 or 2^{19} and LIMIT to 2^{19} or 2^{20} .

The two 40-digit runs produced:

```

LIMIT=524288
DPS=40
checked_nonpowers=524268
failures=0
min_lower_log_margin=(1.4147089634746718e-15, 189427, 231498127)
min_upper_log_margin=(3.0841010256409534e-15, 306982, 497586055)
sha256_m_n_manifest=02b589461fa8c1721bc251575b98b1e3d58001f54cb7faa2cd6c59edb8c819a5
elapsed_seconds=18.505

```

Listing 2: Lower half output.

```

LIMIT=1048576
DPS=40
checked_nonpowers=524287
failures=0
min_lower_log_margin=(9.490232037370244e-16, 958460, 3023921230)
min_upper_log_margin=(2.3019781003173366e-15, 556061, 1275861767)
sha256_m_n_manifest=fccccc5de8673379d37ff0fc840ab091d9ff7a2352ff64a93091f01c21bf4cb4
elapsed_seconds=18.904

```

Listing 3: Upper half output.

Elapsed times are machine-dependent sanity checks; the counts, zero-failure results, extremal tuples, and hashes are the reproducibility data. The hashes cover the ASCII stream `m:n`; of proposed-and-certified pairs and make accidental changes to the candidate sequence detectable across reruns.

References

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